

SingleRAN

X2 Self-Management in SA Networking Feature Parameter Description

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Huawei Technologies Co., Ltd.

Address: Huawei Industrial Base
Bantian, Longgang
Shenzhen 518129
People's Republic of China

Website: <https://www.huawei.com>

Email: support@huawei.com

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1 Change History

The following describes the specific changes in different issues.

SRAN22.1 Draft B (2026-01-30)

This issue includes the following changes.

- Technical changes
None
- Editorial changes
 - Added MML command examples for the **SHARE_FREQ_X2_UP_OPT_SW** option of the **EnodebAlgoExtSwitch.NetworkPrfmOptSwitch** parameter, **X2SON_X2U_AUTO_DEL_OPT_SWITCH** option of the **gNBX2SonConfig.X2SonConfigSwitch** parameter, and **X2_UP_SELF_SETUP_NO_MOCN_SW** option of the **gNBInterfaceParam.InterfaceSelfCfgOptSw** parameter. For details, see [4.7.1.2 Using MML Commands](#).
 - Added a statement that the **SHARE_FREQ_X2_UP_OPT_SW** option of the **EnodebAlgoExtSwitch.NetworkPrfmOptSwitch** parameter can be selected only in RAN sharing with common carrier scenarios. For details, see [4.6.2.1 Prerequisite Functions](#).
 - Revised other descriptions in this document.

SRAN22.1 Draft A (2025-12-31)

This issue is the first release.

2 About This Document

2.1 General Statements

Purpose

This document is intended to acquaint readers with:

- The technical principles of features and their related parameters
- The scenarios where these features are used, the benefits they provide, and the impact they have on networks and functions
- Requirements that must be met before feature activation
- Parameter configuration required for feature activation, verification of feature activation, and monitoring of feature performance

NOTE

This document only provides guidance for feature activation. Feature deployment and feature gains depend on the specifics of the network scenario where the feature is deployed. To achieve optimal gains, contact Huawei professional service engineers.

Functions mentioned in this document work properly only when enabled in the specified applicable scenarios (such as RAT and networking). If a function not mentioned in this document is enabled or a function is enabled in a scenario not specified as applicable, exceptions or other impacts may occur.

Software Interfaces

Any parameters, alarms, counters, or managed objects (MOs) described in this document apply only to the corresponding software release. For future software releases, refer to the corresponding updated product documentation.

2.2 Applicable RAT

This document applies to LTE FDD, LTE TDD, and NR.

For definitions of base stations described in this document, see section "Base Station Products" in *SRAN Networking and Evolution Overview*.

2.3 Features in This Document

This document describes the following features.

RAT	Feature ID	Feature Name	Chapter/Section
LTE FDD	MRFD-131121	LTE and NR X2 Interface Self-Configuration (LTE FDD)	4 X2 Self-Management
LTE TDD	MRFD-131131	LTE and NR X2 Interface Self-Configuration (LTE TDD)	
NR	MRFD-131161	LTE and NR X2 Interface Self-Configuration (NR)	

2.4 Differences

Table 2-1 Differences between LTE FDD and LTE TDD

Function Name	Difference	Chapter/Section
X2 self-management	None	4 X2 Self-Management

Table 2-2 Differences between NR FDD and NR TDD

Function Name	Difference	Chapter/Section
X2 self-management	None	4 X2 Self-Management

Table 2-3 Differences between NSA and SA

Function Name	Difference	Chapter/Section
X2 self-management	Supported only in SA networking	4 X2 Self-Management

Table 2-4 Differences between high frequency bands and low frequency bands

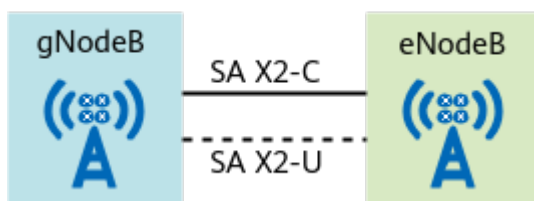
Function Name	Difference	Chapter/Section
X2 self-management	None	4 X2 Self-Management

This document refers to frequency bands belonging to FR1 (410–7125 MHz) as low frequency bands, and those belonging to FR2 (24250–71000 MHz) as high frequency bands. For details about FR1 and FR2, see section 5.1 "General" in 3GPP TS 38.104 V17.8.0.

3 Overview

As services develop, service interactions are increasingly required between gNodeBs in SA networking (referred to as gNodeBs in this document) and eNodeBs. However, conventional NSA X2 interfaces cannot meet the new service requirements. To this end, Huawei defines the X2 interface in SA networking (referred to as the SA X2 interface), as shown in [Figure 3-1](#). The SA X2 interface is a logical interface between an eNodeB and a gNodeB. The SA X2 interface includes the X2 interface on the control plane (X2-C) and X2 interface on the user plane (X2-U). X2-C and X2-U forward control- and user-plane data between an eNodeB and a gNodeB, respectively. The SA X2 control plane is based on the Stream Control Transmission Protocol (SCTP) and the user plane is based on the GPRS Tunneling Protocol-User Plane (GTP-U). Compared with the NSA X2 interface, the SA X2 interface supports only X2 self-management functions, including X2 self-setup, X2 self-update, X2 self-removal, and X2 blacklist and whitelist. For related descriptions about the NSA X2 interface, see *X2 and S1 Self-Management in NSA Networking*.

Figure 3-1 SA X2 interface model



4 X2 Self-Management

4.1 X2 Self-Setup

The X2 interface can be configured only in endpoint mode for an eNodeB or a gNodeB.

NOTE

Endpoint mode is a transmission link configuration mode based on endpoints. When configuring a transmission link, you need to configure endpoint information including the control-plane host and peer and user-plane host and peer. NR does not support the simplified endpoint mode. For details about the simplified endpoint mode of LTE, see *S1 and X2 Self-Management* in eRAN feature documentation.

The user-plane and control-plane hosts of an X2 interface must be configured manually. For details, see [4.7.1.2 Using MML Commands](#). The user-plane and control-plane peers can be configured manually or automatically, which must be consistent between the eNodeB and gNodeB.

- Peers manually configured
In this scenario, **SCTPPEER** and **USERPLANEPEER** MOs must be configured for the peers manually. If an eNodeB or gNodeB has multiple X2 interfaces, multiple peer configurations must be manually added.
- Peers automatically configured
In this scenario, the peers do not need to be configured manually. When inter-base-station coordination services (for example, LTE and NR intelligent carrier shutdown in inter-base-station SA networking, as detailed in *Multi-RAT Coordinated Energy Saving*) are performed for UEs, if the eNodeB detects that the X2 link does not exist or is faulty, it exchanges X2 transport-layer configuration data with the gNodeB through the MAE and obtains the IP address of the gNodeB X2 interface. The local port number (specified by the **SCTPHOST.PN (LTE eNodeB, 5G gNodeB)** parameter) used by the SCTP link of the eNodeB must be the same as that of the gNodeB. Otherwise, SCTP link negotiation will fail.

In an LTE and NR co-MPT scenario, when X2 links are set up between the eNodeB and its neighboring eNodeBs as well as between the gNodeB and its neighboring eNodeBs, the local SCTP port number (specified by the

SCTPHOST.PN (LTE eNodeB, 5G gNodeB) parameter) used by the eNodeB in the co-MPT base station must be the same as that used by the gNodeB.

In SA X2 and NSA X2 mixed networking, the X2 interfaces on the eNodeB in EN-DC and the inter-eNodeB X2 interfaces share X2 interface resources. If the self-setup function with peers automatically configured is enabled but self-removal is disabled for inter-eNodeB X2 interfaces, a large number of redundant X2 links may exist, leading to a resource waste. Therefore, it is recommended that X2 self-removal triggered by link faults and X2 self-removal based on X2 usage be enabled for inter-eNodeB X2 interfaces. If the number of established X2 interfaces has reached the maximum value, EN-DC X2 interfaces cannot be set up. To address this issue, it is recommended that inter-eNodeB or LTE-NR inter-RAT X2 self-removal based on dynamic sharing of X2 specifications be enabled. For details about inter-eNodeB X2 self-removal, see *S1 and X2 Self-Management in eRAN Feature Documentation*.

NOTE

- Maximum number of X2-C interfaces supported on the LTE and NR sides:
 - LTE: UMPTb: 256; UMPTe: 384; UMPTga: 512; UMPTg/UMPTHa: 768
 - NR: UMPTe/UMPTb/UMPTga: 384; UMPTg/UMPTHa: 768
- Only X2 over OSS is supported for X2 self-setup with peers automatically configured between eNodeBs and gNodeBs. For details about X2 over OSS, see *S1 and X2 Self-Management in eRAN Feature Documentation*.
- X2 self-setup on only the eNodeB or gNodeB is not supported.
- For an LTE and NR co-MPT base station, the local IP address of the X2 interface on the eNodeB must differ from that of the X2 interface on the gNodeB.
- In RAN sharing with common carrier mode, to separate X2-U link self-setup/self-removal procedures for different operators, the **SHARE_FREQ_X2_UP_OPT_SW** option of the **EnodebAlgoExtSwitch.NetworkPrfmOptSwitch** parameter on the eNodeB side and the **X2SON_X2U_AUTO_DEL_OPT_SWITCH** option of the **gNBX2SonConfig.X2SonConfigSwitch** parameter and the **X2_UP_SELF_SETUP_NO_MOCN_SW** option of the **gNBInterfaceParam.InterfaceSelfCfgOptSw** parameter on the gNodeB side need to be selected at the same time.

The source base station starts a 12-hour timer after sending an X2 self-setup request to a target base station and will not send a second X2 self-setup request to the target base station before the timer expires. This prevents frequent X2 self-setups between the two base stations.

4.1.1 IPv4/IPv6 Single-Stack Transmission

4.1.1.1 X2 Self-Setup with Peers Manually Configured

An SA X2 interface is called X2-C when it is used for the control plane and is called X2-U for the user plane. Currently, the SA X2 interface and NSA X2 interface share the X2-C. [Figure 4-1](#) shows the relationships between MOs involved in SA X2 self-setup with peers manually configured.

Figure 4-1 Relationships between MOs involved in X2 self-setup with peers manually configured

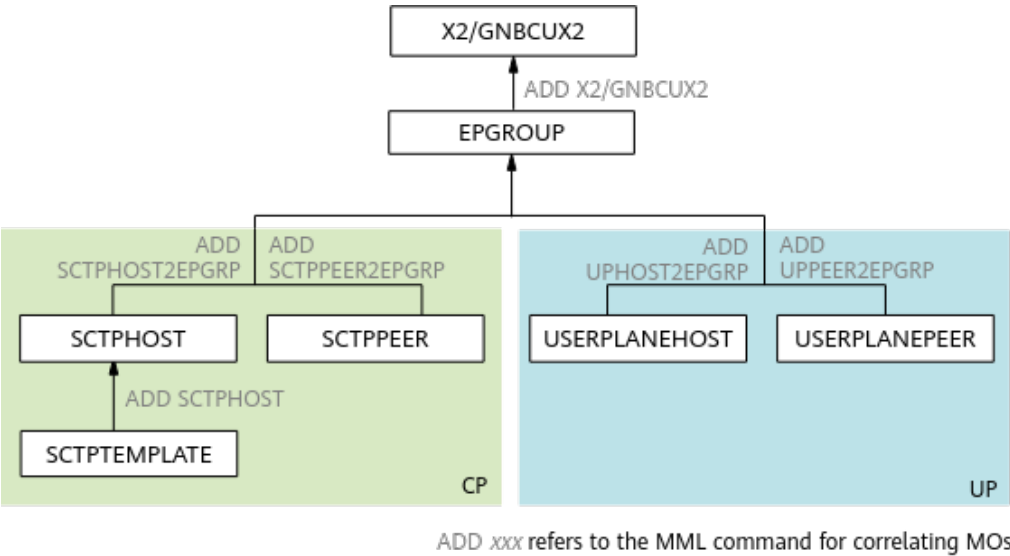


Table 4-1 describes the relationships between the MOs shown in **Figure 4-1**.

Table 4-1 Setting notes for the MOs involved in X2 self-setup with peers manually configured

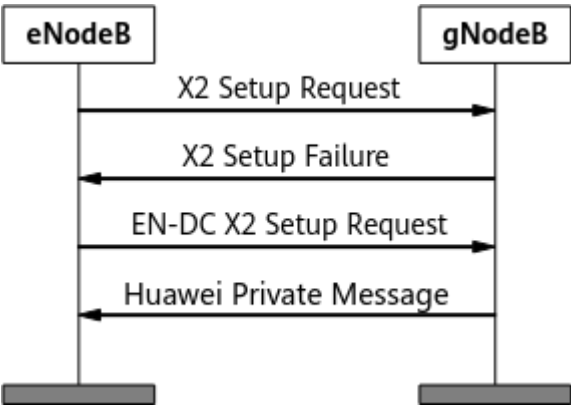
Object	MO at the Local End	MO at the Peer End	Description
Control plane	SCTPHOST NOTE The SCTPHOST.SCTP_TEMPLATEID (LTE eNodeB, 5G gNodeB) and SCTPTEMPLATE.SCTP_TEMPLATEID (LTE eNodeB, 5G gNodeB) parameter values must be the same. In this way, the attribute parameters of the control-plane links are configured consistently with those in the SCTP parameter template.	SCTPPEER	These MOs are used to set parameters such as the local and peer IP addresses and the port numbers of a control-plane link. - The SCTPHOST.PN (LTE eNodeB, 5G gNodeB) and SCTPPEER.PN (LTE eNodeB, 5G gNodeB) parameters must be set to the same value. The recommended value is 36422. For details, see chapter 7 "Transport layer" in 3GPP TS 36.422 V15.1.0. - The first local IP address (specified by the SCTPHOST.SIGIP1V4 (LTE eNodeB, 5G gNodeB) parameter) and the first peer IP address (specified by the SCTPPEER.SIGIP1V4 (LTE eNodeB, 5G gNodeB) parameter) are used as the local and peer IP addresses of the SCTP link, respectively. The eNodeB or gNodeB automatically sets up a multihoming SCTP link if a second IP address is configured in the SCTPHOST or SCTPPEER MO.

Object	MO at the Local End	MO at the Peer End	Description
User plane	USERPLANEHOST	USERPLANEPEER	These MOs are used to set parameters such as the local and peer IP addresses of a user-plane link. The eNodeB or gNodeB automatically sets up a transmission link for the X2-U interface using the local X2 user-plane IP address (specified by the USERPLANEHOST.LOCIPV4 (LTE eNodeB, 5G gNodeB) parameter) and the peer gNodeB or eNodeB IP address (specified by the USERPLANEPEER.PEERIPV4 (LTE eNodeB, 5G gNodeB) parameter). If the USERPLANEPEER MO is configured for the eNodeB, such configuration is not required for the gNodeB and self-setup can be directly performed.
Endpoint group	EPGROUP	EPGROUP	When adding the control-plane host and peer as well as the user-plane host and peer to endpoint groups: • The control-plane host and peer of an X2 interface must be added to the same endpoint group. • The user-plane host and peer of an X2 interface must be added to the same endpoint group.
X2 interface	X2	X2	This MO is used to add an X2 object on the eNodeB and bind it with the endpoint group.
X2 interface	gNBCUX2	gNBCUX2	This MO is used to add an X2 object on the gNodeB and bind it with the endpoint group.

After the preceding configurations are complete, if the **LNR_SA_GNB_X2_SETUP_SW** option of the **eNodeBOamParam.InterfaceSonConfigSwitch** parameter on the LTE side and the **X2_SETUP_FOR_SA_SW** option of the **gNBInterfaceParam.X2ConfigSw** parameter on the NR side are selected, the eNodeB automatically initiates the X2 interface setup procedure.

As the **SCTPPEER** MO is manually configured, the local eNodeB cannot determine whether the peer NE is an eNodeB or a gNodeB. In this case, the local eNodeB sends an X2 Setup Request message to the peer NE, as shown in **Figure 4-2**. After receiving this message, the peer NE responds with an X2 Setup Failure message, with the cause value of "scg-mobility". The eNodeB then determines that the peer NE is a gNodeB, and sends an EN-DC X2 Setup Request message. The gNodeB then responds with a Huawei Private Message to set up a private X2 link.

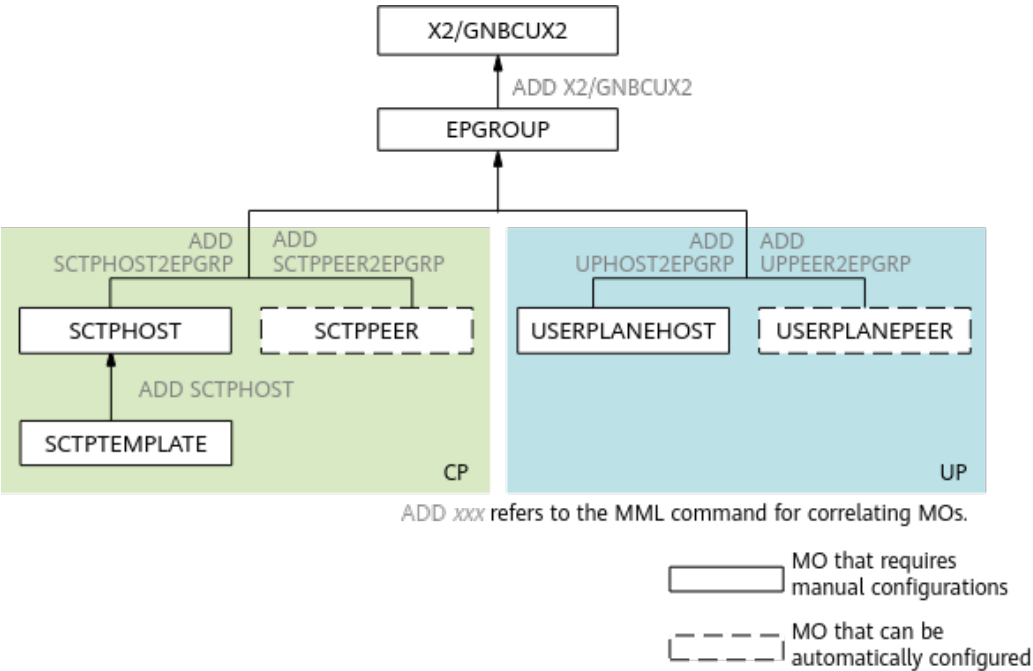
Figure 4-2 Procedure of X2 interface setup with peers manually configured



4.1.1.2 X2 Self-Setup with Peers Automatically Configured

X2 self-setup with peers automatically configured, as shown in **Figure 4-3**, is recommended because it is free from manual operations, reducing costs on network planning, optimization, and O&M.

Figure 4-3 Relationships between MOs involved in X2 self-setup with peers automatically configured



X2 self-setup with peers automatically configured is triggered when inter-base-station coordination services are performed between the eNodeB and gNodeB.

X2 self-setup with peers automatically configured on the eNodeB side requires that (1) the **LTE_NR_X2_SON_SETUP_SW** option of the **GlobalProcSwitch.InterfaceSetupPolicySw** parameter and the **LNR_INTER_BS_PWR_SAVE_SW** option of the **EnodebPwrSavingAlgo.PwrSavingAlgoSwitch** parameter be selected on the eNodeB side; (2) the **X2SON_SETUP_SWITCH** option of the **gNBX2SonConfig.X2SonConfigSwitch** parameter and the **SA_LNR_INTER_BS_PWR_SAVE_SW** option of the **gNBPwrSavingAlgo.PwrSavingAlgoSwitch** parameter be selected on the gNodeB side.

 NOTE

To prevent redundant X2 interfaces from being set up, it is recommended that the **NrExternalCell.NrNetworkingOption** parameter be set to **SA** on the eNodeB when an SA X2 interface is set up between the gNodeB and eNodeB, where the gNodeB does not serve any NSA cell.

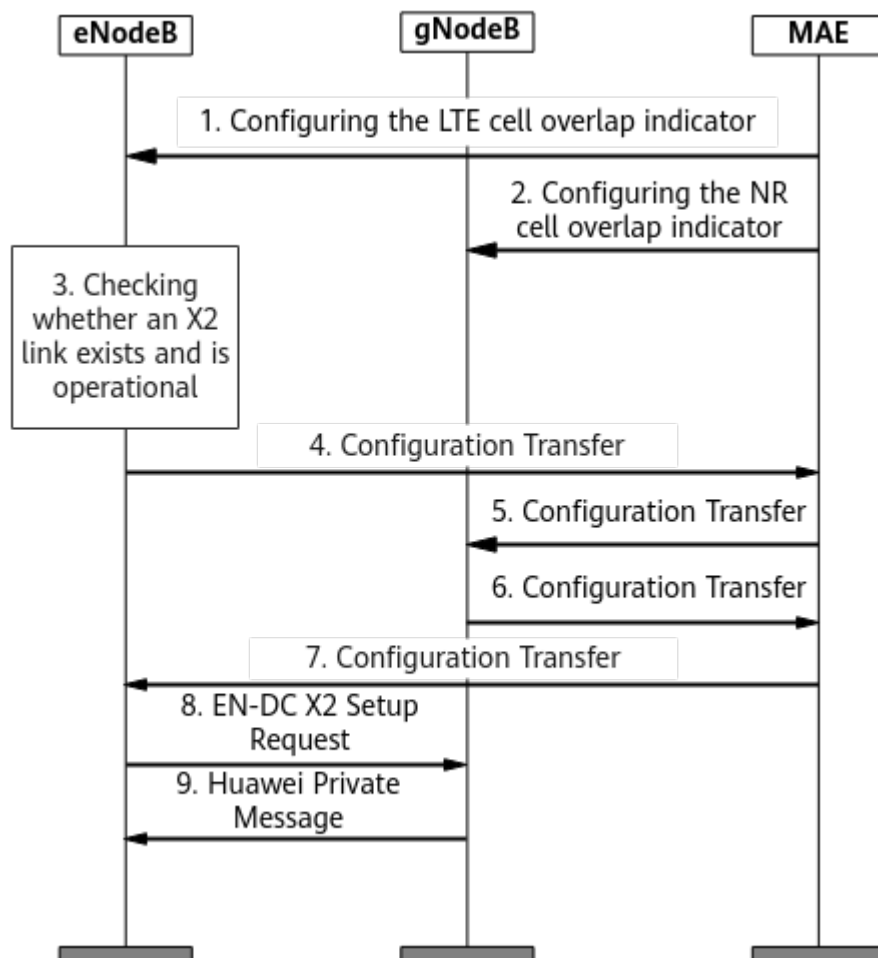
4.1.1.2.1 OSS-based Automatic Peer Configuration for the eNodeB and gNodeB

In this mode, the eNodeB, gNodeB, and MAE must meet the following conditions:

- For the eNodeB
 - The following MOs have been configured: **SCTPHOST**, **USERPLANEHOST**, **X2**, **EPGROUP**, and **SCTPTEMPLATE**.
 - Neighbor relationships with NR cells have been configured.
- For the gNodeB
 - The following MOs have been configured: **SCTPHOST**, **USERPLANEHOST**, **gNBCUX2**, **EPGROUP**, and **SCTPTEMPLATE**.
- For the MAE
 - The CMServer service has been enabled. Otherwise, the MAE cannot respond to the X2 self-setup request from the eNodeB. For details about CMServer, see **Description > Function Description > MAE-Access Function Description > MAE Function Description About Software Management > Processes and Services** in *iMaster MAE Product Documentation (EulerOS, TaiShan)*.
 - The eNodeB and gNodeB are managed by the same MAE.

Currently, the SA X2 self-setup is triggered by inter-base-station coordination services. **Figure 4-4** shows the procedure of X2 self-setup between an eNodeB and a gNodeB, using LTE and NR intelligent carrier shutdown in inter-base-station scenarios in SA networking as an example.

Figure 4-4 X2 self-setup with peers automatically configured for the eNodeB and gNodeB



The procedure is as follows:

1. The MAE configures the overlap indicator (controlled by the **CARRIER_SHUTDOWN_OVERLAP_FLAG** option of the **NrNRRelationship.AggregationAttribute** parameter) for an LTE cell that has a co-coverage relationship with an NR cell.

NOTE

If the co-coverage relationship between the NR and LTE cells is updated based on the LTE and NR co-coverage learning result, the MAE updates the overlap indicator on the LTE side.

2. The MAE configures the overlap indicator (controlled by the **LNR_CARR_SHUTDOWN_OVERLAP_IND** option of the **NRCellEutraNRRelation.PowerSavingOverlapInd** parameter) for an NR cell that has a co-coverage relationship with an LTE cell.
3. The source eNodeB checks whether an X2 link with the target gNodeB exists and is operational for inter-base-station coordination services.
 - If the X2 link exists and is operational, no X2 self-setup procedure will be triggered.

- If the first two options below are not selected on the source eNodeB or the last two options below are not selected on the target gNodeB, the self-setup procedure is not started.
 - **LNR_SA_GNB_X2_SETUP_SW** option of the **eNodeBOamParam.InterfaceSonConfigSwitch** parameter
 - **LNR_INTER_BS_PWR_SAVE_SW** option of the **EnodebPwrSavingAlgo.PwrSavingAlgoSwitch** parameter
 - **X2_SETUP_FOR_SA_SW** option of the **gNBInterfaceParam.X2ConfigSw** parameter
 - **SA_LNR_INTER_BS_PWR_SAVE_SW** option of the **gNBPwrSavingAlgo.PwrSavingAlgoSwitch** parameter
- X2 self-setup is triggered if the following conditions apply: (1) The X2 link is faulty or does not exist; (2) the aforesaid former two options are selected on the source eNodeB; (3) the aforesaid latter two options are selected on the target gNodeB. Then, the procedure proceeds to 4.
- 4. The source eNodeB sends a Configuration Transfer message to the MAE. The message contains the X2-related information (such as the X2-C IP address, X2-U IP address, and operator ID) of the source eNodeB.
- 5. After receiving the message, the MAE forwards it to the target gNodeB according to the target gNodeB ID.

 NOTE

If the MAE fails to find the target gNodeB, the X2 self-setup procedure is terminated as a failure.

6. Upon receiving the message, the target gNodeB sends a Configuration Transfer message carrying its control- and user-plane IP addresses to the MAE. In addition, the target gNodeB automatically generates related MOs to set up transmission links for the control plane and user plane.
 - a. The target gNodeB automatically generates the **USERPLANEPEER** and **SCTPPEER** MOs based on eNodeB's control- and user-plane IP addresses included in this message. The target gNodeB then automatically adds the generated MOs to the **EPGROUP** MO.
 - b. The target gNodeB automatically generates the **gNBCUX2Interface** MO based on manually configured MOs and automatically created MOs. Manually configured MOs include **SCTPHOST**, **USERPLANEHOST**, **gNBCUX2**, **EPGROUP**, and **SCTPTEMPLATE**. Automatically created MOs include **SCTPPEER** and **USERPLANEPEER** generated in a.
7. The MAE forwards the message to the source eNodeB.

Upon receiving the message, the source eNodeB automatically generates related MOs to set up transmission links for the control plane and user plane.

 - a. The source eNodeB automatically generates the **USERPLANEPEER** and **SCTPPEER** MOs based on the control- and user-plane IP addresses of the target end included in this message. The source eNodeB then automatically adds the generated MOs to the **EPGROUP** MO.
 - b. The source eNodeB automatically generates the **X2Interface** MO based on manually configured MOs and automatically created MOs. Manually configured MOs include **SCTPHOST**, **USERPLANEHOST**, **X2**, **EPGROUP**,

and **SCTPTEMPLATE**. Automatically created MOs include **SCTPPEER** and **USERPLANEPEER** generated in [a](#).

 NOTE

If the MAE fails to find the source eNodeB, the X2 self-setup procedure fails.

8. The source eNodeB then sends an EN-DC X2 Setup Request message to the target gNodeB.
9. After receiving the EN-DC X2 Setup Request message, the target gNodeB responds with a Huawei Private Message to set up an SA X2 link.

4.1.2 IPv4/IPv6 Dual-Stack Transmission

If an eNodeB or gNodeB supports both IPv4 and IPv6 transmission, the eNodeB or gNodeB supports the dual-stack function. In single-operator scenarios, the X2 user-plane link and X2 control-plane link must use the same IP version. In RAN sharing scenarios, the X2 user-plane link and X2 control-plane link can use different IP versions if they belong to different operators. Either of the following methods can be used to establish an X2 link between the eNodeB and the gNodeB, depending on their configured IP versions:

- X2 self-setup with peers manually configured

The local and peer ends for an X2 link must use the same IP version (either IPv4 or IPv6). For example, if the local and peer ends use IPv4, the X2 link uses IPv4 as the protocol stack. For details about how to manually configure peers, see [4.1.1.1 X2 Self-Setup with Peers Manually Configured](#).

 NOTE

In this manual configuration mode, both IPv4 and IPv6 are configured. In this way, there is an attempt to establish two X2 links between local and peer ends. The second X2 link fails to be set up, but double transmission link resources are occupied. Therefore, in this mode, you are advised not to configure both IPv4 and IPv6 at the same time.

- X2 self-setup with peers automatically configured

Based on the local IP stack (IPv4/IPv6) and the IP version preference (specified by **EPGROUP.IPVERPREFERENCE (5G gNodeB, LTE eNodeB)**) configured on the eNodeB and gNodeB, X2 links with different IP versions are automatically established between them, as described in [Table 4-2](#). For details about how to automatically configure peers, see [4.1.1.2 X2 Self-Setup with Peers Automatically Configured](#).

If the IPv4/IPv6 dual-stack transmission is used over the X2 interfaces between base stations, the X2 interfaces must use the same IP version, which is specified by the **EPGROUP.IPVERPREFERENCE (5G gNodeB, LTE eNodeB)** parameter.

Table 4-2 Rules of configuring the IP version for the X2 link between the eNodeB and the gNodeB

IP Version for the eNodeB	IP Version for the gNodeB	IP Version Preference for the gNodeB	IP Version for the X2 Link
IPv4	IPv4	N/A	IPv4
IPv6	IPv6	N/A	IPv6
IPv4/IPv6	IPv4	N/A	IPv4
IPv4/IPv6	IPv6	N/A	IPv6
IPv4	IPv4/IPv6	N/A	IPv4
IPv6	IPv4/IPv6	N/A	IPv6
IPv4/IPv6	IPv4/IPv6	IPv4	IPv4
IPv4/IPv6	IPv4/IPv6	IPv6	IPv6

 NOTE

- A change in the **EPGROUP.IPVERPREFERENCE (5G gNodeB, LTE eNodeB)** parameter setting will not cause the control-plane or user-plane X2 link to reestablish. The parameter setting will be checked only when X2 link setup or reestablishment is required.
- If both IPv4 and IPv6 addresses are configured on the eNodeB for which automatic peer configuration is used, the eNodeB sends two IP addresses to the peer gNodeB. If the peer gNodeB is configured with only one IP address, the IP version of this IP address is used for X2 link setup. If the peer gNodeB is configured with both IPv4 and IPv6 addresses, the IP version used for X2 link setup is determined by the **EPGROUP.IPVERPREFERENCE (5G gNodeB, LTE eNodeB)** parameter setting on the gNodeB.

4.2 X2 Self-Update

X2 self-update of peer information can be service-triggered, or it can be triggered manually.

4.2.1 Service-Triggered Self-Update of X2 Peer Information

During network operation, **SCTPPEER** and **USERPLANEPEER** MOs of the X2 interface will be automatically updated if all the following conditions apply: (1) The IP address or operator information of the peer is changed, such as the mobile network code (MNC), mobile country code (MCC), or global eNodeB/gNodeB ID; (2) the X2 interface is faulty; (3) X2 self-setup is triggered by inter-base-station coordination services.

Change in the IP Address of the Peer

If the control-plane IP address of the peer is changed, the X2 interface with the peer will become faulty. If an inter-base-station coordination service is initiated

between the eNodeB and gNodeB when the X2 interface is faulty, X2 self-setup will be triggered. The local end obtains the new control-plane IP address of the peer during the self-setup. The local end searches for an **SCTPPEER** MO that contains the same operator information (indicated by **SCTPPEER.REMOTEID (LTE eNodeB, 5G gNodeB)**) as the operator information (such as MNC, MCC, and global eNodeB/gNodeB ID) in the Configuration Transfer message. The local end then updates the control-plane IP address in the **SCTPPEER** MO.

If the user-plane IP address of the peer is changed, the X2 interface with the peer will become faulty. If an inter-base-station coordination service is initiated between the eNodeB and gNodeB when the X2 interface is faulty, X2 self-setup will be triggered. The local end obtains the new user-plane IP address of the peer during the self-setup. The local end searches for a **USERPLANEPEER** MO that contains the same operator information (indicated by **SCTPPEER.REMOTEID (LTE eNodeB, 5G gNodeB)**) as the operator information (MNC, MCC, and global eNodeB/gNodeB ID) in the Configuration Transfer message. The local end then generates a new **USERPLANEPEER** MO based on the modified user-plane IP address of the peer. The faulty user-plane link leads to the reporting of ALM-25954 User Plane Fault and is deleted through X2 self-removal (see [4.3 X2 Self-Removal](#) for details).

NOTE

Based on information about the local and peer endpoints included in the **EPGROUP** MO, the eNodeB or gNodeB can automatically generate the **SCTPLNK** and **CPBEARER** MOs with the **SCTPLNK.CTRLMODE (LTE eNodeB, 5G gNodeB)** and **CPBEARER.CTRLMODE (LTE eNodeB, 5G gNodeB)** parameters set to **AUTO_MODE**. Do not manually modify or remove these automatically generated MOs. Otherwise, the corresponding X2 interface cannot work properly. If an X2 interface fails to work properly due to the preceding reason, remove the **X2** MO and then add it again.

Change in the Operator Information of the Peer

If the operator information (such as MNC, MCC, and global gNodeB ID) of the peer is changed, the X2 interface does not become faulty, and the values of the **SCTPPEER.REMOTEID (LTE eNodeB, 5G gNodeB)** and **USERPLANEPEER.REMOTEID (LTE eNodeB, 5G gNodeB)** parameters do not automatically update since X2 self-setup is not triggered.

X2 self-setup is triggered only when inter-base-station coordination services are initiated again in the case that the X2 interface is faulty due to a transmission link fault or any other reason. During this process, the local end searches for peers (specified by the **SCTPPEER** and **USERPLANEPEER** MOs) that match the peer IP address information contained in the Configuration Transfer message. The local end then updates the values of the corresponding **SCTPPEER.REMOTEID (LTE eNodeB, 5G gNodeB)** and **USERPLANEPEER.REMOTEID (LTE eNodeB, 5G gNodeB)** parameters.

NOTE

If the switch for inter-base-station coordination services is turned on again, X2 self-setup will not be triggered within 30 minutes.

Change in the IP Address and Operator Information of the Peer

If both the IP address and operator information of the peer on the X2 interface are changed, the local end cannot obtain the peer information, and therefore does not automatically update the **SCTPPEER** or **USERPLANEPEER** MO.

4.2.2 Manually-Triggered Self-Update of X2 Peer Information

For the X2 interface, the IP addresses of the local base station are referenced in the **SCTPPEER** and **USERPLANEPEER** MOs of the peer base station. When the IP addresses of the local base station are changed, the MAE can automatically associate and change the **SCTPPEER** and **USERPLANEPEER** MOs of the peer base station that has the same routing domain (**VRF** MO) as the local base station.

4.3 X2 Self-Removal

X2 Self-Removal Triggered by Link Faults

An X2 interface is automatically removed when any of the following conditions is met:

- If the **EnodebAlgoExtSwitch.X2SonDelTimerForLNX2Fault** parameter is set to a value other than **0** on the LTE side, a timer specified by this parameter starts when an X2 control-plane link fault occurs. If the fault persists until the timer expires, local X2 configurations, including the settings of the **SCTPPEER** and **USERPLANEPEER** MOs, are removed. The timer starts 10 minutes after the local base station starts up. This prevents the X2 interface from being mistakenly removed when the timer expires during the local base station startup.

It is recommended that the

EnodebAlgoExtSwitch.X2SonDelTimerForLNX2Fault parameter be set to a value specifying a period of at least 10 minutes. If this period is extremely short, the X2 interface will be mistakenly removed.

- If an X2 control-plane link fault occurs after the **X2SON_DEL_FOR_X2FAULT_SWITCH** option of the **gNBX2SonConfig.X2SonConfigSwitch** parameter is selected on the NR side, an exception timer starts. If the fault persists throughout the period specified by the **gNBX2SonConfig.X2SonDeleteTimerForX2Fault** parameter on the NR side, local X2 configurations, including the settings of the **SCTPPEER** and **USERPLANEPEER** MOs, are removed.

X2 Self-Removal Triggered by Immediate Faults After Initial X2 Self-Setup

An X2 interface is automatically removed when all the following conditions are met:

- The X2 interface is initially set up, and the SCTP link is faulty.
- On the LTE and NR sides, the corresponding switches are turned on:
 - The **EnodebAlgoExtSwitch.GnbX2InitFailDelSwitch** parameter is set to **ON** on the LTE side. The eNodeB identifies a link that has a fault immediately after initial X2 self-setup is completed. After the fault lasts for several minutes (less than 10 minutes), the eNodeB automatically

removes and blacklists this X2 link. For details, see [4.4.2 Dynamic X2 Blacklist](#).

- The **X2INIT_FAIL_DEL_SWITCH** option of the **gNBX2SonConfig.X2SonConfigSwitch** parameter is selected on the NR side. The gNodeB identifies a link that has a fault immediately after initial X2 self-setup is completed. After the fault lasts for several minutes (less than 10 minutes), the gNodeB automatically removes this link.
- The **SCTPPEER** MO is automatically configured.

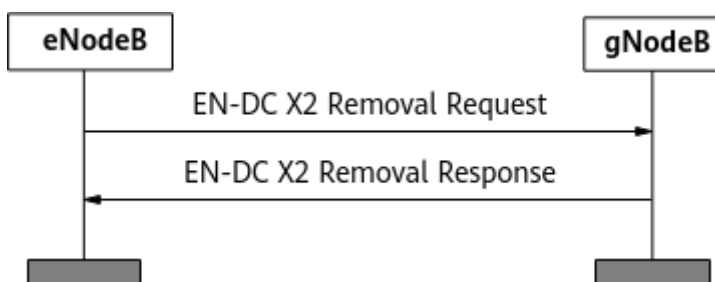
When the preceding conditions are met:

- If the **SCTPPEER.CTRLMODE** parameter is set to **AUTO_MODE**, the X2 interface and the **SCTPLNK** and **SCTPPEER** MOs are removed.
- If the **SCTPPEER.CTRLMODE** parameter is set to **MANUAL_MODE**, the X2 interface and the **SCTPLNK** MO are removed. The **SCTPPEER** MO is not removed.

X2 Self-Removal Based on X2 Usage

- On the LTE side, when the **EnodebAlgoExtSwitch.X2SonDelTimerForLNX2Usage** parameter is set to a non-zero value, the X2 self-removal based on X2 usage function is triggered within a duration specified by this parameter if no inter-base-station coordination service is performed over an X2 interface. In this case, local X2 configurations, including the settings of the **SCTPPEER** and **USERPLANEPEER** MOs, are removed. When the **SCTPPEER.CTRLMODE** and **USERPLANEPEER.CTRLMODE** parameters are set to **AUTO_MODE**, automatically created MOs (such as the **X2Interface** MO) are also removed.

Figure 4-5 LTE-triggered X2 self-removal based on X2 usage

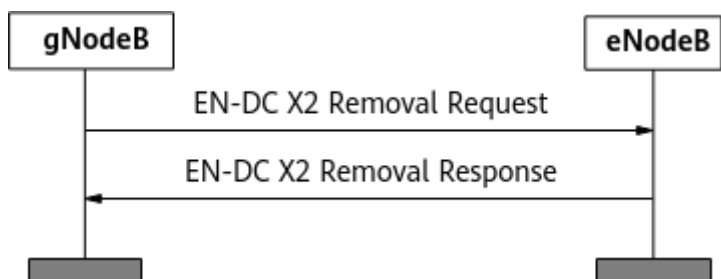


In **Figure 4-5**:

- a. The eNodeB sends an EN-DC X2 Removal Request message to the peer gNodeB, instructing the gNodeB to remove its X2 interface. The gNodeB sends an EN-DC X2 Removal Response message and removes its X2 interface.
 - b. After receiving the EN-DC X2 Removal Response message, the eNodeB removes its X2 interface.
- On the NR side, when the **X2SON_DEL_FOR_X2USAGE_SWITCH** option of the **gNBX2SonConfig.X2SonConfigSwitch** parameter is selected, the X2 self-removal based on X2 usage function is triggered within a duration specified by the **gNBX2SonConfig.X2SonDeleteTimerForX2Usage** parameter if no inter-base-station coordination service is performed over an X2 interface. In

this case, local X2 configurations, including the settings of the **SCTPPEER** and **USERPLANEPEER** MOs, are removed. When the **SCTPPEER.CTRLMODE** and **USERPLANEPEER.CTRLMODE** parameters are set to **AUTO_MODE**, automatically created MOs (such as the **X2Interface** MO) are also removed.

Figure 4-6 NR-triggered X2 self-removal based on X2 usage



In **Figure 4-6**:

- The gNodeB sends an EN-DC X2 Removal Request message to the peer eNodeB, instructing the eNodeB to remove its X2 interface. The eNodeB sends an EN-DC X2 Removal Response message and removes its X2 interface.
- After receiving the EN-DC X2 Removal Response message, the gNodeB removes its X2 interface.

NOTE

- Before enabling the X2 self-removal function on the LTE or NR side, ensure that the X2 self-setup switch is turned on. Otherwise, inter-base-station coordination services may fail.
- The remote ID of the X2 interface to be removed must comply with the following format: "eNB:MCC_xxx MNC_xxx GlobaleNBId_xxx-xxx-xxxxxxx" (for an eNodeB) or "gNB:MCC_xxx MNC_xxx GlobalgNBId_xxx-xxx-xxxxxxx" (for a gNodeB).
- A smaller value of the **gNBX2SonConfig.X2SonDeleteTimerForX2Usage** parameter results in a higher probability of meeting the conditions for X2 link removal. As a result, X2 links are quickly removed and then automatically set up, causing frequent X2 link additions and removals. The period must be longer than at least one day when fast and automatic adjustment of the X2 interface is required.

X2 Self-Removal Based on Percentage of Capacity Reserved for High-Priority X2 Interfaces on the LTE Side

When the **LNR_SA_GNB_X2_SETUP_SW** option of the **eNodeBOamParam.InterfaceSonConfigSwitch** parameter is selected on the eNodeB, if the number of available X2 links on the eNodeB is less than the X2 link capacity supported by the main control board multiplied by the **GlobalProcSwitch.HighPriX2RsvdPct** parameter value, the eNodeB no longer initiates self-setup of low-priority X2 links and triggers self-removal of SA X2 links after high-priority X2 links are successfully set up.

X2-U Peer Self-Removal

When the **USERPLANEPEER.CTRLMODE** parameter is set to **AUTO_MODE** on the eNodeB or gNodeB side, the eNodeB or gNodeB can automatically remove the X2-U transmission links and the **USERPLANEPEER** MOs. When this parameter is set

to **MANUAL_MODE**, the eNodeB or gNodeB does not automatically remove the X2-U transmission links and the **USERPLANEPEER** MOs.

When the **GEPMODELPARA.UPAUTODELSW (LTE eNodeB, 5G gNodeB)** parameter is set to **ENABLE**, the eNodeB or gNodeB determines whether to automatically remove the X2-U transmission links in either of the following situations:

- Self-removal based on link faults
If the GTP-U detection result indicates that all the transmission links corresponding to **USERPLANEPEER** MOs are faulty, the eNodeB or gNodeB periodically (every 60 minutes) attempts to remove those transmission links (up to 50 links at one time) that remain faulty for over one hour and the corresponding **USERPLANEPEER** MOs.
- Self-removal based on the aging mechanism
 - In case of an excess over the specifications of a base station
The eNodeB or gNodeB marks a transmission link as aged and processes as follows if the link has not been used by any service.
 - The base station performs self-removal based on the aging mechanism once every hour and checks whether the number of established transmission links reaches or exceeds 90% of the maximum number of transmission links supported. If it does, the base station removes aged transmission links (up to 50 links at one time) and the corresponding **USERPLANEPEER** MOs. If it does not, no aged transmission links are removed.
 - When new transmission links need to be established, the base station checks whether the number of established transmission links reaches or exceeds 90% of the maximum number of transmission links supported. If it does, the base station removes aged transmission links (up to 50 links at one time) and the corresponding **USERPLANEPEER** MOs, and establishes new transmission links. If it does not, no aged transmission links are removed.
 - If the maximum allowed number of transmission links is reached and there is no aged link, the eNodeB or gNodeB rejects new UEs' access requests.
 - In case of an excess over the specifications of a single endpoint group
"Self-removal in case of an excess over the specifications of a single endpoint group" takes effect for the eNodeB or gNodeB when the **SINGLE_EPUP_AUTO_DEL_SW** option of the **TRANSALGEXTPARA.TNLBEARERMNGALGSW (5G gNodeB, LTE eNodeB)** parameter is selected. Self-removal based on the aging mechanism starts when the number of peers in a single endpoint group exceeds 90% of the maximum number allowed by the in-use board. The condition for marking a link as aged and the self-removal mechanism are the same as those for self-removal in case of an excess over the specifications of a base station.

 NOTE

When the user planes of the SA X2 and Xn interfaces or those of the SA X2 and eXn interfaces use the same IP address, to ensure that services on the Xn or eXn interface are not affected, the peer of the X2-U interface is not removed when X2 self-removal is triggered.

4.4 X2 Blacklist and Whitelist

4.4.1 Static X2 Blacklist and Whitelist

Operators can run the **ADD ENDCX2BLACKWHITELIST** command on the eNodeB to add a peer gNodeB to an EN-DC X2 blacklist or whitelist.

If a peer gNodeB is on the EN-DC X2 blacklist, an X2 link between the eNodeB and this gNodeB cannot be automatically set up. If the peer gNodeB that has set up an X2 link with the eNodeB is added to the EN-DC X2 blacklist, the X2 link will become faulty and cannot be automatically removed. It is recommended that this X2 link be manually removed.

If a peer gNodeB is on the EN-DC X2 whitelist, an X2 link between the eNodeB and this gNodeB cannot be automatically removed.

 NOTE

If the ID of a blacklisted or whitelisted gNodeB is changed, you need to manually update the value of the **EnDcX2BlackWhiteList.gNodeBId** parameter. Otherwise, if the X2 link set up with this gNodeB is not removed, the blacklist or whitelist function still takes effect for the gNodeB.

A maximum of 128 EN-DC X2 blacklist and whitelist entries can be set.

4.4.2 Dynamic X2 Blacklist

An X2 interface cannot be automatically set up when the local and peer base stations of the X2 interface are in the dynamic blacklist and the gNodeB X2 dynamic blacklist aging timer (specified by the **EnodebAlgoExtSwitch.GnbX2DynBlklistAgingTimer** parameter) has not expired. Specifically, the local eNodeB neither initiates a self-setup request for the X2 interface nor responds to the X2 self-setup request initiated by the peer gNodeB. When the timer expires, the local eNodeB removes the peer gNodeB from the corresponding blacklist.

The dynamic X2 blacklist specification is the same as the X2 interface specification. The specification is shared by LTE-LTE X2 and LTE-NR X2 interfaces. In the multi-MPT co-BBU scenario, the dynamic X2 blacklist specification is the same as the maximum X2 interface specification among main control boards. If the number of entries in a dynamic blacklist has reached the maximum and a new entry needs to be added, the new entry will replace the one with the shortest remaining aging time.

Operators can run the **CLR X2DYNBLACKLIST** command on the LTE side to remove a peer gNodeB from the dynamic X2 blacklist before the gNodeB X2 dynamic blacklist aging timer expires.

4.5 Network Analysis

4.5.1 Benefits

SA X2 self-management simplifies configuration operations, reduces operating expense (OPEX) for operators, improves the eNodeB/gNodeB resource usage rate, and enables inter-base-station coordination services more efficiently.

4.5.2 Impacts

The impacts between this function and other functions are described in [4.6.2.3 Function Impacts](#).

- When the mis-deletion prevention mechanism for X2 self-removal triggered by link faults is in effect, the number of X2 interface setup attempts at the local eNodeB (indicated by the **L.Sig.X2.SendSetup.Att** counter) and the number of successful X2 interface setups at the local eNodeB (indicated by the **L.Sig.X2.SendSetup.Succ** counter) may decrease.
- SA X2 link setup attempts may cause the following alarms to be reported:
 - ALM-29801 gNodeB X2 Interface Full Specification
 - ALM-29225 Maximum Number of X2 Interfaces Reached
 - ALM-29800 gNodeB X2 Interface Fault
 - ALM-29204 X2 Interface Fault
- SA X2 link setup attempts may cause the values of the counters related to the following function subsets to fluctuate: SCTP Link Measurement, eNodeB Transmission Interface Measurement, and X2 Link Measurement. For descriptions of the preceding counters, see *3900 & 5900 Series Base Station Performance Counter Reference* in *3900 & 5900 Series Base Station Product Documentation*.
- SA X2 link setups occupy the X2 specifications of the main control board. If the X2 specifications are insufficient, the counters related to NSA handovers and number of UEs may be affected. For details about the impacts, see "Impacts" in *X2 and S1 Self-Management in NSA Networking*.

4.6 Requirements

4.6.1 Licenses

None

4.6.2 Functions

Before activating this function, ensure that its prerequisite functions have been activated and mutually exclusive functions have been deactivated, if there are any. For detailed operations, see the relevant feature documents.

Additionally, understand any potential function impacts before activating this function. To deactivate any impacted functions, follow the operations described in the relevant feature documents.

4.6.2.1 Prerequisite Functions

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD NR FDD Low-frequency NR TDD High-frequency NR TDD	Automatic user plane removal	GEPMODELPA RA.UPAUTODE LSW (LTE eNodeB, 5G gNodeB)	<i>X2 and S1 Self- Management in NSA Networking</i>	Ensure that the GEPMODELPA.RA.UPAUTODELSW (LTE eNodeB, 5G gNodeB) parameter is set to ENABLE before enabling "self-removal in case of an excess over the specifications of a single endpoint group" (by selecting the SINGLE_EPUP_AUTO_DEL_SW option of the TRANSALGEXTPARA.TNLBEARERMNGALGSW (LTE eNodeB, 5G gNodeB) parameter).
LTE FDD LTE TDD NR FDD Low-frequency NR TDD High-frequency NR TDD	Self-removal in case of an excess over the specifications of a single endpoint group	SINGLE_EPUP_ AUTO_DEL_S W option of the TRANSALGEXTPARA.TNLBEARERMNGALGSW (LTE eNodeB, 5G gNodeB) parameter	<i>X2 and S1 Self- Management in NSA Networking</i>	Ensure that "self-removal in case of an excess over the specifications of a single endpoint group" is disabled before disabling automatic user plane removal (by setting the GEPMODELPA.RA.UPAUTODELSW (LTE eNodeB, 5G gNodeB) parameter to DISABLE).

RAT	Function Name	Function Switch	Reference	Description
LTE FDD LTE TDD	RAN sharing with common carrier	ENodeBSharingMode.ENodeBSharingMode set to SHARED_FREQ	<i>RAN Sharing in eRAN Feature Documentation</i>	The SHARE_FREQ_X2_UP_OPT_SW option of the EnodebAlgoExtSwitch.NetworkPrfmOptSwitch parameter can be selected only when the ENodeBSharingMode.ENodeBSharingMode parameter is set to SHARED_FREQ .

4.6.2.2 Mutually Exclusive Functions

RAT	Function Name	Function Switch	Reference	Description
NR FDD Low-frequency NR TDD High-frequency NR TDD	Operator-specific NCI configuration	gNBSharingMode.MultipleNCISwitch	<i>Multi-Operator Sharing</i>	The SA X2 setup switch must be turned off when operator-specific NCI configuration is enabled.

4.6.2.3 Function Impacts

None

4.6.3 Hardware

For details about the compatibility between the hardware described in this section, see the technical specifications and hardware description of the corresponding hardware in *3900 & 5900 Series Base Station Product Documentation*.

Base Station Models

On the LTE side, the compatible base stations are as follows:

- 3900 and 5900 series base stations (macro base stations). 5900 series base stations must be configured with the BBU5900, BBU5900A, BBU5910, BookBBU5901a, or BookBBU5901.
- DBS3900 LampSite and DBS5900 LampSite

On the NR side, the compatible base stations are as follows:

- 3900 and 5900 series base stations (macro base stations). 3900 series base stations must be configured with the BBU3910, and 5900 series base stations must be configured with the BBU5900, BBU5900A, BBU5910, BookBBU5901a, or BookBBU5901.
- DBS3900 LampSite and DBS5900 LampSite. DBS3900 LampSite must be configured with the BBU3910.

Boards

- Main control boards: UMPTb and later main control boards
- Baseband processing units: UBBPe and later baseband processing units

RF Modules

N/A

4.6.4 Networking

The gNodeB serving an NR cell does not serve any NSA cell.

4.6.5 Others

- Huawei base stations are required at both ends of an SA X2 interface, and the transport network must be functional. (For example, if IPv4 single-stack transmission is used for LTE, IPv4 single-stack or IPv4/IPv6 dual-stack transmission needs to be used for NR.)
- The eNodeB and gNodeB on an SA X2 link must be managed by the same MAE.

4.7 Operation and Maintenance

4.7.1 Data Configuration

4.7.1.1 Data Preparation

Configuring Common Parameters

Common parameters must be configured at both the local and peer ends.

The following table describes the parameters that must be set in an **SCTPTEMPLATE** MO.

Parameter Name	Parameter ID	Setting Notes
SCTP Parameters Template ID	SCTPTEMPLATE.SCTPTE MPLATEID (5G gNodeB, LTE eNodeB)	Set this parameter based on the network plan.
DSCP Switch	SCTPTEMPLATE.DSCPSW (LTE eNodeB, 5G gNodeB)	The default value is recommended.
Switch Back Flag	SCTPTEMPLATE.SWITCH BACKFLAG (LTE eNodeB, 5G gNodeB)	Set this parameter based on the network plan.

The following table describes the parameters that must be set in an **SCTPHOST** MO.

Parameter Name	Parameter ID	Setting Notes
SCTP Host ID	SCTPHOST.SCTPHOSTID (5G gNodeB, LTE eNodeB)	Set this parameter based on the network plan.
IP Version	SCTPHOST.IPVERSION (5G gNodeB, LTE eNodeB)	Set this parameter based on the network plan.
First Local IP Address	SCTPHOST.SIGIP1V4 (5G gNodeB, LTE eNodeB)	Set this parameter based on the network plan.
First Local IPv6 Address	SCTPHOST.SIGIP1V6 (5G gNodeB, LTE eNodeB)	Set this parameter based on the network plan.
Second Local IP Address	SCTPHOST.SIGIP2V4 (5G gNodeB, LTE eNodeB)	Set this parameter based on the network plan.
Second Local IPv6 Address	SCTPHOST.SIGIP2V6 (5G gNodeB, LTE eNodeB)	Set this parameter based on the network plan.
Local SCTP Port No.	SCTPHOST.PN (5G gNodeB, LTE eNodeB)	Set this parameter based on the network plan.
SCTP Parameters Template ID	SCTPHOST.SCTPTEMPLA TEID (5G gNodeB, LTE eNodeB)	Set this parameter based on the network plan.
IPSec Auto Configure Switch for First Local IP	SCTPHOST.SIGIP1SECSW ITCH (LTE eNodeB, 5G gNodeB)	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
IPSec Auto Configure Switch for Second Local IP	SCTPHOST.SIGIP2SECSWITCH (LTE eNodeB, 5G gNodeB)	Set this parameter based on the network plan.

(Optional, required for manual peer configuration) The following table describes the parameters that must be set in an **SCTPPEER** MO.

Parameter Name	Parameter ID	Setting Notes
IP Version	SCTPPEER.IPVERSION (5G gNodeB, LTE eNodeB)	Set this parameter based on the network plan.
First Peer IP Address	SCTPPEER.SIGIP1V4 (5G gNodeB, LTE eNodeB)	Set this parameter based on the network plan.
First Peer IPv6 Address	SCTPPEER.SIGIP1V6 (5G gNodeB, LTE eNodeB)	Set this parameter based on the network plan.
Second Peer IP Address	SCTPPEER.SIGIP2V4 (5G gNodeB, LTE eNodeB)	Set this parameter based on the network plan.
Second Peer IPv6 Address	SCTPPEER.SIGIP2V6 (5G gNodeB, LTE eNodeB)	Set this parameter based on the network plan.
Peer SCTP Port No.	SCTPPEER.PN (5G gNodeB, LTE eNodeB)	Set this parameter based on the network plan.

The following table describes the parameters that must be set in a **USERPLANEHOST** MO.

Parameter Name	Parameter ID	Setting Notes
User Plane Host ID	USERPLANEHOST.UPHOSTID (5G gNodeB, LTE eNodeB)	Set this parameter based on the network plan.
IP Version	USERPLANEHOST.IPVERSION (5G gNodeB, LTE eNodeB)	• If the peer uses IPv4, set this parameter to IPv4. • If the peer uses IPv6, set this parameter to IPv6.

Parameter Name	Parameter ID	Setting Notes
Local IP Address	USERPLANEHOST.LOCIP V4 (5G gNodeB, LTE eNodeB)	Set this parameter based on the network plan.
Local IPv6 Address	USERPLANEHOST.LOCIP V6 (5G gNodeB, LTE eNodeB)	Set this parameter based on the network plan.
IPSec Auto Configure Switch	USERPLANEHOST.IPSECS WITCH (LTE eNodeB, 5G gNodeB)	Set this parameter based on the network plan.
Master Flag	USERPLANEHOST.FLAG (LTE eNodeB, 5G gNodeB)	Set this parameter based on the network plan.

(Optional, required for manual peer configuration) The following table describes the parameters that must be set in a **USERPLANEPEER** MO.

Parameter Name	Parameter ID	Setting Notes
User Plane Peer ID	USERPLANEPEER.UPPEERID (5G gNodeB, LTE eNodeB)	Set this parameter based on the network plan.
IP Version	USERPLANEPEER.IPVERSION (5G gNodeB, LTE eNodeB)	• If the peer uses IPv4, set this parameter to IPv4. • If the peer uses IPv6, set this parameter to IPv6.
Peer IP Address	USERPLANEPEER.PEERIP V4 (5G gNodeB, LTE eNodeB)	Set this parameter based on the network plan.
Peer IPv6 Address	USERPLANEPEER.PEERIP V6 (5G gNodeB, LTE eNodeB)	Set this parameter based on the network plan.
IPSec Auto Configure Switch	USERPLANEPEER.IPSECS WITCH (LTE eNodeB, 5G gNodeB)	Set this parameter based on the network plan.

The following table describes the parameters that must be set in an **EPGROUP** MO. It is recommended that the **USERPLANEHOST** and **USERPLANEPEER** MOs or the **SCTPHOST** and **SCTPPEER** MOs for an operator be configured in the same **EPGROUP** MO.

Parameter Name	Parameter ID	Setting Notes
End Point Group ID	EPGROUP.EPGROUPID (5G gNodeB, LTE eNodeB)	Set this parameter based on the network plan.
SCTP Host List	EPGROUP.SCTPHOSTLIST (5G gNodeB, LTE eNodeB)	Set this parameter based on the network plan.
SCTP Peer List	EPGROUP.SCTPPEERLIST (5G gNodeB, LTE eNodeB)	Set this parameter based on the network plan. If there are multiple peers, multiple SCTP peers exist in the list specified by the EPGROUP.SCTPPEERLIST (5G gNodeB, LTE eNodeB) parameter.
User Plane Host List	EPGROUP.USERPLANEHOSTLIST (5G gNodeB, LTE eNodeB)	Set this parameter based on the network plan.
User Plane Peer List	EPGROUP.USERPLANEPEERLIST (5G gNodeB, LTE eNodeB)	Set this parameter based on the network plan.
IP Protocol Version Preference	EPGROUP.IPVERPREFERENCE (5G gNodeB, LTE eNodeB)	Set this parameter based on the network plan.

Configuring Parameters on the LTE Side

(Required for automatic SA X2 configuration) The following table describes how to configure the **LNR_SA_GNB_X2_SETUP_SW** and **LNR_INTER_BS_PWR_SAVE_SW** options using SA X2 link setup triggered by LTE and NR intelligent carrier shutdown in inter-base-station SA networking as an example.

Parameter Name	Parameter ID	Option	Setting Notes
Interface SON Configuration Switch	eNodeBOamParam.InterfaceSonConfigSwitch	LNR_SA_GNB_X2_SETUP_SW	Select this option.
Power Saving Algorithm Switch	EnodebPwrSavingAlgorithm.PwrSavingAlgorithm	LNR_INTER_BS_PWR_SAVE_SW	Select this option.

(Required for manual SA X2 configuration) The following table describes how to configure the **LNR_SA_GNB_X2_SETUP_SW** option.

Parameter Name	Parameter ID	Option	Setting Notes
Interface SON Configuration Switch	eNodeBOamParam.InterfaceSonConfigSwitch	LNR_SA_GNB_X2_SETUP_SW	Select this option.

The following table describes the parameters that must be set in the **GlobalProcSwitch** and **EnodebAlgoExtSwitch** MOs.

Parameter Name	Parameter ID	Option	Setting Notes
Interface Setup Policy Switch	GlobalProcSwitch.InterfaceSetupPolicySw	LTE_NR_X2_SON_SETUP_SW	Set this parameter based on the network plan.
X2 Setup Failure Num Threshold	GlobalProcSwitch.X2SetupFailureNumThd	N/A	Set this parameter based on the network plan.
X2 Setup Failure Num Timer	GlobalProcSwitch.X2SetupFailureNumTimer	N/A	Set this parameter based on the network plan.
gNodeB X2 Initial Fail Delete Switch	EnodebAlgoExtSwitch.GnbX2InitFailDelSwitch	N/A	Set this parameter based on the network plan.
X2 SON Delete Timer for LTE-NR X2 Fault	EnodebAlgoExtSwitch.X2SonDelTimerForLNX2Fault	N/A	Set this parameter based on the network plan. It is recommended that this parameter be set to a value greater than or equal to 10 if X2 self-removal triggered by link faults is required.
X2 SON Delete Timer for LTE-NR X2 Usage	EnodebAlgoExtSwitch.X2SonDelTimerForLNX2Usage	N/A	Set this parameter based on the network plan.
gNodeB X2 Dynamic Blacklist Aging Timer	EnodebAlgoExtSwitch.GnbX2DynBlklistAgingTimer	N/A	Set this parameter based on the network plan.

The following table describes the parameters that must be set to configure the reference relationship between the **X2** and **EPGROUP** MOs.

Parameter Name	Parameter ID	Setting Notes
X2 ID	<i>X2.X2Id</i>	Set this parameter based on the network plan.
CN Operator ID	<i>X2.CnOperatorId</i>	Set this parameter based on the network plan.
End Point Group Config Flag	<i>X2.EpGroupCfgFlag</i>	This parameter specifies whether to configure the control-plane or user-plane endpoint group. - Set this parameter to CP_CFG when only the control-plane EPGROUP MO is referenced by the X2 MO. - Set this parameter to UP_CFG when only the user-plane EPGROUP MO is referenced by the X2 MO. - Set this parameter to CP_UP_CFG when both control-plane and user-plane EPGROUP MOs are referenced by the X2 MO.
Control Plane End Point Group ID	<i>X2.CpEpGroupId</i>	Set this parameter based on the network plan.
User Plane End Point Group ID	<i>X2.UpEpGroupId</i>	Set this parameter based on the network plan.
Peer Base Station Release	<i>X2.TargetENodeBRelease</i>	Set this parameter based on the network plan.

The following table describes the parameters that must be set in an **EnDcX2BlackWhiteList** MO on the eNodeB side.

Parameter Name	Parameter ID	Setting Notes
Mobile country code	EnDcX2BlackWhiteList.Mcc	Set this parameter based on the network plan.
Mobile network code	EnDcX2BlackWhiteList.Mnc	Set this parameter based on the network plan.

Parameter Name	Parameter ID	Setting Notes
gNodeB ID	EnDcX2BlackWhiteList.gNodeBId	Set this parameter based on the network plan.
List Type	EnDcX2BlackWhiteList.ListType	Set this parameter based on the network plan.

The following table describes the parameters that must be set in an **SCTPHOST** MO.

Parameter Name	Parameter ID	Setting Notes
Simple Mode Switch	SCTPHOST.SIMPLEMODESWITCH (LTE eNodeB, 5G gNodeB)	Set this parameter based on the network plan.

The following table describes the parameters that must be set in an **EnodebAlgoExtSwitch** MO.

Parameter Name	Parameter ID	Setting Notes
Simple Mode Switch	EnodebAlgoExtSwitch.NetworkPrfmOptSwitch	It is recommended that the SHARE_FREQ_X2_UP_OPT_SW option be selected if X2-U link self-setup/self-removal procedures need to be separated for different operators in SA networking.

Configuring Parameters on the NR Side

(Required for automatic SA X2 configuration) The following table describes how to configure the **X2_SETUP_FOR_SA_SW** and **SA_LNR_INTER_BS_PWR_SAVE_SW** options using SA X2 link setup triggered by LTE and NR intelligent carrier shutdown in inter-base-station SA networking as an example.

Parameter Name	Parameter ID	Option	Setting Notes
X2 Config Switch	gNBInterfaceParam.X2ConfigSw	X2_SETUP_FOR_SA_SW	Select this option.

Parameter Name	Parameter ID	Option	Setting Notes
Power Saving Algorithm Switch	gNBPwrSavingAlgo.PwrSavingAlgoSwitch	SA_LNR_INTER_BS_PWR_SAV E_SW	Select this option.

(Required for manual SA X2 configuration) The following table describes how to configure the **X2_SETUP_FOR_SA_SW** option.

Parameter Name	Parameter ID	Option	Setting Notes
X2 Config Switch	gNBInterfaceParam.X2ConfigSw	X2_SETUP_FOR_SA_SW	Select this option.

The following table describes the parameters that must be set in a **gNBX2SonConfig** MO.

Parameter Name	Parameter ID	Option	Setting Notes
X2 SON Configuration Switch	gNBX2SonConfig.X2SonConfigSwitch	X2SON_SETUP_SWITCH	Set this parameter based on the network plan.
X2 SON Configuration Switch	gNBX2SonConfig.X2SonConfigSwitch	X2INIT_FAIL_DEL_SWITCH	Set this parameter based on the network plan.
X2 SON Configuration Switch	gNBX2SonConfig.X2SonConfigSwitch	X2SON_DEL_FOR_X2FAULT_SWITCH	Set this parameter based on the network plan. gNBX2SonConfig.X2SonDeleteTimerForX2Fault specifies the timer for X2 self-removal based on link faults.

Parameter Name	Parameter ID	Option	Setting Notes
X2 SON Configuration Switch	gNBX2SonConfig.X2SonConfigSwitch	X2SON_DEL_F OR_X2USAGE_ SWITCH	Set this parameter based on the network plan. gNBX2SonConfig.X2SonDeleteGnbAddCntThld specifies the threshold for X2 self-removal based on X2 usage.
X2 SON Delete Timer for X2 Fault	gNBX2SonConfig.X2SonDeleteTimerForX2Fault	N/A	Set this parameter based on the network plan. This is a timer for X2 self-removal based on link faults.
X2 SON Delete Timer for X2 Usage	gNBX2SonConfig.X2SonDeleteTimerForX2Usage	N/A	Set this parameter based on the network plan.

The following table describes the parameters that must be set to configure the reference relationship between the **gNBCUX2** and **EPGROUP** MOs.

Parameter Name	Parameter ID	Setting Notes
gNodeB CU X2 ID	gNBCUX2.gNBCuX2Id	Set this parameter based on the network plan.
Control Plane Endpoint Group ID	gNBCUX2.CpEpGroupId	Set this parameter based on the network plan.
User Plane Endpoint Group ID	gNBCUX2.UpEpGroupId	Set this parameter based on the network plan.

The following table describes the parameters that must be set in the **EnodebAlgoExtSwitch** and **gNBInterfaceParam** MOs.

Parameter Name	Parameter ID	Setting Notes
X2 SON Configuration Switch	gNBX2SonConfig.X2SonConfigSwitch	It is recommended that the X2SON_X2U_AUTO_DEL_OPT_SWITCH option be selected if X2-U link self-setup procedures need to be separated for different operators in SA networking.
Interface Self-Config Optimization Switch	gNBInterfaceParam.InterfaceSelfConfigOptSw	It is recommended that the X2_UP_SELF_SETUP_NO_MOCN_SW option be selected if X2-U link self-removal procedures need to be separated for different operators in SA networking.

If the SON function is used to automatically configure, update, and remove S1 and X2 interfaces, the **GTRANSPARA.EPOBJAUTOCFGSW** parameter must be set to its default value **ENABLE**.

4.7.1.2 Using MML Commands

Before using MML commands, refer to [4.6 Requirements](#) and complete the parameter configurations for related functions based on the impact, dependency, and mutually exclusive relationships between the functions, as well as the actual network scenario. In addition, ensure that the license, hardware, networking, and cell requirements are met.

Before using MML commands, confirm the related parameter modification precautions by referring to the fields including "Service Interrupted After Modification" and "Caution" for the corresponding parameters in the parameter reference.

X2 Self-Setup (Automatic Peer Configuration)

Configuring parameters on the eNodeB (IPv4/IPv6 single-stack transmission)

```
//Using SA X2 link setup triggered by LTE and NR intelligent carrier shutdown in inter-base-station SA
networking as an example
//Turning on the LNR SA gNodeB X2 link setup switch
MOD ENODEBOAMPARAM: InterfaceSonConfigSwitch=LNR_SA_GNB_X2_SETUP_SW-1;
//Turning on the LNR inter-base-station power saving switch
MOD ENODEBPWRSAVINGALGO: PwrSavingAlgoSwitch=LNR_INTER_BS_PWR_SAVE_SW-1;
//Configuring common parameters
//(Optional) Adding an SCTP template. By default, the eNodeB has an SCTP template with
SCTPTEMPLATEID set to 0.
ADD SCTPTEMPLATE: SCTPTEMPLATEID=0, SWITCHBACKFLAG=ENABLE, DSCPSW=OFF;
//Adding an endpoint group for the X2 interface (in IPv4 networking)
ADD EPGROUP: EPGROUPID=1;
//Adding an endpoint group for the X2 interface (in IPv6 networking)
ADD EPGROUP: EPGROUPID=1;
//Adding an X2 control-plane host (in IPv4 networking)
ADD SCTPHOST: SCTPHOSTID=1, IPVERSION=IPv4, SIGIP1V4="x.x.x.x", SIGIP1SECSWITCH=DISABLE,
SIGIP2SECSWITCH=DISABLE, PN=36422, SCTPTEMPLATEID=0;
//Adding an X2 control-plane host (in IPv6 networking)
```

```
ADD SCTPHOST: SCTPHOSTID=1, IPVERSION=IPv6, SIGIP1V6="xx::xx:xx:xx", SIGIP1SECSWITCH=DISABLE,
SIGIP2SECSWITCH=DISABLE, PN=36422, SCTPTemplateID=0;
//Adding an X2 user-plane host (in IPv4 networking)
ADD USERPLANEHOST: UPHOSTID=1, IPVERSION=IPv4, LOCIPV4="x.x.x.x", IPSECSWITCH=DISABLE;
//Adding an X2 user-plane host (in IPv6 networking)
ADD USERPLANEHOST: UPHOSTID=1, IPVERSION=IPv6, LOCIPV6="xx::xx:xx:xx", IPSECSWITCH=DISABLE;
//Adding the X2 user-plane host to the endpoint group
ADD UPHOST2EPGRP: EPGROUPID=1, UPHOSTID=1;
//Adding the X2 control-plane host to the endpoint group
ADD SCTPHOST2EPGRP: EPGROUPID=1, SCTPHOSTID=1;
//Adding an X2 object on the eNodeB side
ADD X2: X2Id=0, CnOperatorId=0, EpGroupCfgFlag=CP_UP_CFG, CpEpGroupId=1, UpEpGroupId=1;
//Enabling X2 self-setup on the eNodeB side
MOD GLOBALPROCSWITCH: InterfaceSetupPolicySw=LTE_NR_X2_SON_SETUP_SW-1;
//Besides preceding X2 configurations, the following X2 configurations are required in dual-MPT networking
//Old model
//(Optional, not required if a default route has been configured) Adding a route from the eNodeB to the
gNodeB (The two IP addresses of the gNodeB are in the network segment of the route.)
ADD IPRT: RTIDX=0, SN=7, SBT=BASE_BOARD, DSTIP="x.x.x.x", DSTMASK="x.x.x.x", RTTYPE=NEXTHOP,
NEXTHOP="x.x.x.x";
//New model
//(Optional, not required if a default route has been configured) Adding a route from the eNodeB to the
gNodeB (The two IP addresses of the gNodeB are in the network segment of the route.)
ADD IPROUTE4: RTIDX=0, DSTIP="x.x.x.x", DSTMASK="x.x.x.x", RTTYPE=NEXTHOP, NEXTHOP="x.x.x.x";
```

Configuring parameters on the eNodeB (IPv4/IPv6 dual-stack transmission)

```
//Using SA X2 link setup triggered by LTE and NR intelligent carrier shutdown in inter-base-station SA
networking as an example
//Turning on the LNR SA gNodeB X2 link setup switch
MOD ENODEBOAMPARAM: InterfaceSonConfigSwitch=LNR_SA_GNB_X2_SETUP_SW-1;
//Turning on the LNR inter-base-station power saving switch
MOD ENODEBPWRSAVINGALGO: PwrSavingAlgoSwitch=LNR_INTER_BS_PWR_SAVE_SW-1;
//Configuring common parameters
//(Optional) Adding an SCTP template. By default, the eNodeB has an SCTP template with
SCTPTemplateID set to 0.
ADD SCTPTemplate: SCTPTemplateID=0, SWITCHBACKFLAG=ENABLE, DSCPSW=OFF;
//Adding an endpoint group for the X2 interface
ADD EPGROUP: EPGROUPID=1, IPVERPREFERENCE=IPv6;
//Adding X2 control-plane hosts
ADD SCTPHOST: SCTPHOSTID=1, IPVERSION=IPv4, SIGIP1V4="x.x.x.x", SIGIP1SECSWITCH=DISABLE,
SIGIP2SECSWITCH=DISABLE, PN=36422, SCTPTemplateID=0;
ADD SCTPHOST: SCTPHOSTID=2, IPVERSION=IPv6, SIGIP1V6="xx::xx:xx:xx", PN=36422, SCTPTemplateID=0;
//Adding X2 user-plane hosts
ADD USERPLANEHOST: UPHOSTID=1, IPVERSION=IPv4, LOCIPV4="x.x.x.x", IPSECSWITCH=DISABLE;
ADD USERPLANEHOST: UPHOSTID=2, IPVERSION=IPv6, LOCIPV6="xx::xx:xx:xx", IPSECSWITCH=DISABLE;
//Adding the X2 user-plane hosts to the endpoint group
ADD UPHOST2EPGRP: EPGROUPID=1, UPHOSTID=1;
ADD UPHOST2EPGRP: EPGROUPID=1, UPHOSTID=2;
//Adding the X2 control-plane hosts to the endpoint group
ADD SCTPHOST2EPGRP: EPGROUPID=1, SCTPHOSTID=1;
ADD SCTPHOST2EPGRP: EPGROUPID=1, SCTPHOSTID=2;
//Adding an X2 object on the eNodeB side
ADD X2: X2Id=0, CnOperatorId=0, EpGroupCfgFlag=CP_UP_CFG, CpEpGroupId=1, UpEpGroupId=1;
//Enabling X2 self-setup on the eNodeB side
MOD GLOBALPROCSWITCH: InterfaceSetupPolicySw=LTE_NR_X2_SON_SETUP_SW-1;
```

Configuring parameters on the gNodeB (IPv4/IPv6 single-stack transmission)

```
//Using SA X2 link setup triggered by LTE and NR intelligent carrier shutdown in inter-base-station SA
networking as an example
//Turning on the SA X2 setup switch
MOD GNBINTERFACEPARAM: X2ConfigSw=X2_SETUP_FOR_SA_SW-1;
//Turning on the SA LNR inter-base-station power saving switch
MOD GNBPWRSAVINGALGO: PwrSavingAlgoSwitch=SA_LNR_INTER_BS_PWR_SAVE_SW-1;
//Configuring common parameters
//(Optional) Adding an SCTP template. By default, the gNodeB has an SCTP template with
SCTPTemplateID set to 0.
ADD SCTPTemplate: SCTPTemplateID=0, SWITCHBACKFLAG=ENABLE, DSCPSW=OFF;
//Adding an endpoint group for the X2 interface (in IPv4 networking)
```

```

ADD EPGROUP: EPGROUPID=1;
//Adding an endpoint group for the X2 interface (in IPv6 networking)
ADD EPGROUP: EPGROUPID=1;
//Adding an X2 control-plane host (in IPv4 networking)
ADD SCTPHOST: SCTPHOSTID=1, IPVERSION=IPv4, SIGIP1V4="x.x.x.x", SIGIP1SECSWITCH=DISABLE,
SIGIP2SECSWITCH=DISABLE, PN=36422, SCTPTEMPLATEID=0;
//Adding an X2 control-plane host (in IPv6 networking)
ADD SCTPHOST: SCTPHOSTID=1, IPVERSION=IPv6, SIGIP1V6="xx:xx:xx:xx", SIGIP1SECSWITCH=DISABLE,
SIGIP2SECSWITCH=DISABLE, PN=36422, SCTPTEMPLATEID=0;
//Adding an X2 user-plane host (in IPv4 networking)
ADD USERPLANEHOST: UPHOSTID=1, IPVERSION=IPv4, LOCIPV4="x.x.x.x", IPSECSWITCH=DISABLE;
//Adding an X2 user-plane host (in IPv6 networking)
ADD USERPLANEHOST: UPHOSTID=1, IPVERSION=IPv6, LOCIPV6="xx:xx:xx:xx", IPSECSWITCH=DISABLE;
//Adding the X2 user-plane host to the endpoint group
ADD UPHOST2EPGRP: EPGROUPID=1, UPHOSTID=1;
//Adding the X2 control-plane host to the endpoint group
ADD SCTPHOST2EPGRP: EPGROUPID=1, SCTPHOSTID=1;
//Adding an X2 object on the gNodeB side
ADD GNBCUX2: gNBCuX2Id=1, CpEpGroupId=1, UpEpGroupId=1;
//Enabling X2 self-setup on the gNodeB side
MOD GNBX2SONCONFIG: X2SonConfigSwitch=X2SON_SETUP_SWITCH-1;
//Besides preceding X2 configurations, the following X2 configurations are required in dual-MPT networking
//Old model
//Configuring a source IP route for each of the two local IP addresses of the X2 interface by using the local
IP address as the source IP address (The two IP addresses of the gNodeB are in the network segment of the
route.)
ADD SRCIPRT: SRCRTIDX=0, SN=6, SBT=BASE_BOARD, SRCIP="x.x.x.x", RTTYPE=NEXTHOP,
NEXTHOP="x.x.x.x";
ADD SRCIPRT: SRCRTIDX=1, SN=7, SBT=BASE_BOARD, SRCIP="x.x.x.x", RTTYPE=NEXTHOP,
NEXTHOP="x.x.x.x";
//New model
//Configuring a source IP route for each of the two local IP addresses of the X2 interface by using the local
IP address as the source IP address (The two IP addresses of the gNodeB are in the network segment of the
route.)
ADD SRCIPROUTE4: SRCRTIDX=0, SRCIP="x.x.x.x", RTTYPE=NEXTHOP, NEXTHOP="x.x.x.x";
ADD SRCIPROUTE4: SRCRTIDX=1, SRCIP="x.x.x.x", RTTYPE=NEXTHOP, NEXTHOP="x.x.x.x";
//Adding a control-plane host for the X2 interface (Two local IP addresses must be configured for the SCTP
host.)
ADD SCTPHOST: SCTPHOSTID=0, IPVERSION=IPv4, SIGIP1V4="x.x.x.x", SIGIP1SECSWITCH=DISABLE,
SIGIP2V4="x.x.x.x", SIGIP2SECSWITCH=DISABLE, PN=36422, SIMPLEMODESWITCH=SIMPLE_MODE_OFF,
SCTPTEMPLATEID=0;
//Adding user-plane hosts for the X2 interface (Two user-plane hosts must be configured.)
ADD USERPLANEHOST: UPHOSTID=1, IPVERSION=IPv4, LOCIPV4="x.x.x.x", IPSECSWITCH=DISABLE,
FLAG=MASTER;
ADD USERPLANEHOST: UPHOSTID=10, IPVERSION=IPv4, LOCIPV4="x.x.x.x", IPSECSWITCH=DISABLE,
FLAG=MASTER;

//Adding the two X2 user-plane hosts to the endpoint group
ADD UPHOST2EPGRP: EPGROUPID=1, UPHOSTID=1;
ADD UPHOST2EPGRP: EPGROUPID=1, UPHOSTID=10;

```

Configuring parameters on the gNodeB (IPv4/IPv6 dual-stack transmission)

```

//Using SA X2 link setup triggered by LTE and NR intelligent carrier shutdown in inter-base-station SA
networking as an example
//Turning on the SA X2 setup switch
MOD GNBINTERFACEPARAM: X2ConfigSw=X2_SETUP_FOR_SA_SW-1;
//Turning on the SA LNR inter-base-station power saving switch
MOD GNBPWRSAVINGALGO: PwrSavingAlgoSwitch=SA_LNR_INTER_BS_PWR_SAVE_SW-1;
//Configuring common parameters
//Optional Adding an SCTP template. By default, the gNodeB has an SCTP template with
SCTPTEMPLATEID set to 0.
ADD SCTPTEMPLATE: SCTPTEMPLATEID=0, SWITCHBACKFLAG=ENABLE;
//Adding an endpoint group for the X2 interface
ADD EPGROUP: EPGROUPID=1, IPVERPREFERENCE=IPv6;
//Adding X2 control-plane hosts
ADD SCTPHOST: SCTPHOSTID=1, IPVERSION=IPv4, SIGIP1V4="x.x.x.x", SIGIP1SECSWITCH=DISABLE,
SIGIP2SECSWITCH=DISABLE, PN=36422, SCTPTEMPLATEID=0;
ADD SCTPHOST: SCTPHOSTID=2, IPVERSION=IPv6, SIGIP1V6="xx:xx:xx:xx", SIGIP1SECSWITCH=DISABLE,
SIGIP2SECSWITCH=DISABLE, PN=36422, SCTPTEMPLATEID=0;

```

```
//Adding X2 user-plane hosts
ADD USERPLANEHOST: UPHOSTID=1, IPVERSION=IPv4, LOCIPV4="x.x.x.x", IPSECSWITCH=DISABLE;
ADD USERPLANEHOST: UPHOSTID=2, IPVERSION=IPv6, LOCIPV6="xx:xx:xx:xx", IPSECSWITCH=DISABLE;
//Adding the X2 user-plane hosts to the endpoint group
ADD UPHOST2EPGRP: EPGROUPID=1, UPHOSTID=1;
ADD UPHOST2EPGRP: EPGROUPID=1, UPHOSTID=2;
//Adding the X2 control-plane hosts to the endpoint group
ADD SCTPHOST2EPGRP: EPGROUPID=1, SCTPHOSTID=1;
ADD SCTPHOST2EPGRP: EPGROUPID=1, SCTPHOSTID=2;
//Adding an X2 object on the gNodeB side
ADD GNBCUX2: gNBCuX2Id=1, CpEpGroupId=1, UpEpGroupId=1;
//Enabling X2 self-setup on the gNodeB side
MOD GNBX2SONCONFIG: X2SonConfigSwitch=X2SON_SETUP_SWITCH-1;
```

X2 Self-Setup (Manual Peer Configuration)

Configuring parameters on the eNodeB (IPv4/IPv6 single-stack transmission)

```
//Turning on the LNR SA gNodeB X2 link setup switch
MOD ENODEBOAMPARAM: InterfaceSonConfigSwitch=LNR_SA_GNB_X2_SETUP_SW-1;
//Configuring common parameters
//((Optional) Adding an SCTP template. By default, the eNodeB has an SCTP template with
SCTPTEMPLATEID set to 0.
ADD SCTPTEMPLATE: SCTPTEMPLATEID=0, SWITCHBACKFLAG=ENABLE;
//Adding an endpoint group for the X2 interface
ADD EPGROUP: EPGROUPID=1;
//Adding an X2 control-plane host (in IPv4 networking)
ADD SCTPHOST: SCTPHOSTID=1, IPVERSION=IPv4, SIGIP1V4="x.x.x.x", SIGIP1SECSWITCH=DISABLE,
SIGIP2SECSWITCH=DISABLE, PN=36422, SCTPTEMPLATEID=0;
//Adding an X2 control-plane host (in IPv6 networking)
ADD SCTPHOST: SCTPHOSTID=1, IPVERSION=IPv6, SIGIP1V6="xx:xx:xx:xx", SIGIP1SECSWITCH=DISABLE,
SIGIP2SECSWITCH=DISABLE, PN=36422, SCTPTEMPLATEID=0;
//Adding an X2 control-plane peer (in IPv4 networking)
ADD SCTPPEER: SCTPPEERID=1, IPVERSION=IPv4, SIGIP1V4="x.x.x.x", PN=36422;
//Adding an X2 control-plane peer (in IPv6 networking)
ADD SCTPPEER: SCTPPEERID=1, IPVERSION=IPv6, SIGIP1V6="xx:xx:xx:xx", PN=36422;
//Adding an X2 user-plane host (in IPv4 networking)
ADD USERPLANEHOST: UPHOSTID=1, IPVERSION=IPv4, LOCIPV4="x.x.x.x", IPSECSWITCH=DISABLE;
//Adding an X2 user-plane host (in IPv6 networking)
ADD USERPLANEHOST: UPHOSTID=1, IPVERSION=IPv6, LOCIPV6="xx:xx:xx:xx", IPSECSWITCH=DISABLE;
//Adding an X2 user-plane peer (in IPv4 networking)
ADD USERPLANEPEER: UPPEERID=1, IPVERSION=IPv4, PEERIPV4="x.x.x.x", IPSECSWITCH=DISABLE;
//Adding an X2 user-plane peer (in IPv6 networking)
ADD USERPLANEPEER: UPPEERID=1, IPVERSION=IPv6, PEERIPV6="xx:xx:xx:xx", IPSECSWITCH=DISABLE;
//Adding the X2 user-plane host to the endpoint group
ADD UPHOST2EPGRP: EPGROUPID=1, UPHOSTID=1;
//Adding the X2 user-plane peer to the endpoint group
ADD UPPEER2EPGRP: EPGROUPID=1, UPPEERID=1;
//Adding the X2 control-plane host to the endpoint group
ADD SCTPHOST2EPGRP: EPGROUPID=1, SCTPHOSTID=1;
//Adding the X2 control-plane peer to the endpoint group
ADD SCTPPEER2EPGRP: EPGROUPID=1, SCTPPEERID=1;
//Adding an X2 object on the eNodeB side
ADD X2: X2Id=0, CnOperatorId=0, EpGroupCfgFlag=CP_UP_CFG, CpEpGroupId=1, UpEpGroupId=1;
//Besides preceding X2 configurations, the following X2 configurations are required in dual-MPT networking
//Old model
//((Optional, not required if a default route has been configured) Adding a route from the eNodeB to the
gNodeB (The two IP addresses of the gNodeB are in the network segment of the route.)
ADD IPRT: RTIDX=0, SN=7, SBT=BASE_BOARD, DSTIP="x.x.x.x", DSTMASK="x.x.x.x", RTTYPE=NEXTHOP,
NEXTHOP="x.x.x.x";
//New model
//((Optional, not required if a default route has been configured) Adding a route from the eNodeB to the
gNodeB (The two IP addresses of the gNodeB are in the network segment of the route.)
ADD IPROUTE4: RTIDX=0, DSTIP="x.x.x.x", DSTMASK="x.x.x.x", RTTYPE=NEXTHOP, NEXTHOP="x.x.x.x";
//Adding a control-plane peer for the X2 interface (Two peer IP addresses must be configured for the SCTP
peer.)
ADD SCTPPEER: SCTPPEERID=1, IPVERSION=IPv4, SIGIP1V4="x.x.x.x", SIGIP2V4="x.x.x.x", PN=36422;
//Adding user-plane peers for the X2 interface (Two user-plane peers must be configured.)
ADD USERPLANEPEER: UPPEERID=1, IPVERSION=IPv4, PEERIPV4="x.x.x.x", IPSECSWITCH=DISABLE;
```

```
ADD USERPLANEPEER: UPPEERID=2, IPVERSION=IPv4, PEERIPV4="x.x.x.x", IPSECSWITCH=DISABLE;  
//Adding the two X2 user-plane peers to the endpoint group  
ADD UPPEER2EPGRP: EPGROUPID=1, UPPEERID=1;  
ADD UPPEER2EPGRP: EPGROUPID=1, UPPEERID=2;  
//Adding the SCTP peer of the X2 interface to the endpoint group  
ADD SCTPPEER2EPGRP: EPGROUPID=1, SCTPPEERID=1;
```

Configuring parameters on the eNodeB (IPv4/IPv6 dual-stack transmission)

```
//Turning on the LNR SA gNodeB X2 link setup switch  
MOD ENODEBOAMPARAM: InterfaceSonConfigSwitch=LNR_SA_GNB_X2_SETUP_SW-1;  
//Configuring common parameters  
//(Optional) Adding an SCTP template. By default, the eNodeB has an SCTP template with  
SCTPTEMPLATEID set to 0.  
ADD SCTPTEMPLATE: SCTPTEMPLATEID=0, SWITCHBACKFLAG=ENABLE;  
//Adding an endpoint group for the X2 interface  
ADD EPGROUP: EPGROUPID=1, IPVERPREFERENCE=IPv6;  
//Adding X2 control-plane hosts  
ADD SCTPHOST: SCTPHOSTID=1, IPVERSION=IPv4, SIGIP1V4="x.x.x.x", SIGIP1SECSWITCH=DISABLE,  
SIGIP2SECSWITCH=DISABLE, PN=36422, SCTPTEMPLATEID=0;  
ADD SCTPHOST: SCTPHOSTID=2, IPVERSION=IPv6, SIGIP1V6="xx:xx:xx:xx", PN=36422, SCTPTEMPLATEID=0;  
//Adding an X2 control-plane peer through either of the following configurations  
ADD SCTPPEER: SCTPPEERID=1, IPVERSION=IPv4, SIGIP1V4="x.x.x.x", PN=36422;  
ADD SCTPPEER: SCTPPEERID=2, IPVERSION=IPv6, SIGIP1V6="xx:xx:xx:xx", PN=36422;  
//Adding X2 user-plane hosts  
ADD USERPLANEHOST: UPHOSTID=1, IPVERSION=IPv4, LOCIPV4="x.x.x.x", IPSECSWITCH=DISABLE;  
ADD USERPLANEHOST: UPHOSTID=2, IPVERSION=IPv6, LOCIPV6="xx:xx:xx:xx", IPSECSWITCH=DISABLE;  
//Adding an X2 user-plane peer through either of the following configurations (The IP version on the user  
plane must be the same as that on the control plane.)  
ADD USERPLANEPEER: UPPEERID=1, IPVERSION=IPv4, PEERIPV4="x.x.x.x", IPSECSWITCH=DISABLE;  
ADD USERPLANEPEER: UPPEERID=2, IPVERSION=IPv6, PEERIPV6="xx:xx:xx:xx", IPSECSWITCH=DISABLE;  
//Adding the X2 user-plane hosts to the endpoint group  
ADD UPHOST2EPGRP: EPGROUPID=1, UPHOSTID=1;  
ADD UPHOST2EPGRP: EPGROUPID=1, UPHOSTID=2;  
//Adding the X2 user-plane peers to the endpoint group  
ADD UPPEER2EPGRP: EPGROUPID=1, UPPEERID=1;  
ADD UPPEER2EPGRP: EPGROUPID=1, UPPEERID=2;  
//Adding the X2 control-plane hosts to the endpoint group  
ADD SCTPHOST2EPGRP: EPGROUPID=1, SCTPHOSTID=1;  
ADD SCTPHOST2EPGRP: EPGROUPID=1, SCTPHOSTID=2;  
//Adding the X2 control-plane peers to the endpoint group  
ADD SCTPPEER2EPGRP: EPGROUPID=1, SCTPPEERID=1;  
ADD SCTPPEER2EPGRP: EPGROUPID=1, SCTPPEERID=2;  
//Adding an X2 object on the eNodeB side  
ADD X2: X2Id=0, CnOperatorId=0, EpGroupCfgFlag=CP_UP_CFG, CpEpGroupId=1, UpEpGroupId=1;
```

Configuring parameters on the gNodeB (IPv4/IPv6 single-stack transmission)

```
//Turning on the SA X2 setup switch  
MOD GNBINTERFACEPARAM: X2ConfigSw=X2_SETUP_FOR_SA_SW-1;  
//Configuring common parameters  
//(Optional) Adding an SCTP template. By default, the gNodeB has an SCTP template with  
SCTPTEMPLATEID set to 0.  
ADD SCTPTEMPLATE: SCTPTEMPLATEID=0, SWITCHBACKFLAG=ENABLE;  
//Adding an endpoint group for the X2 interface  
ADD EPGROUP: EPGROUPID=1;  
//Adding an X2 control-plane host (in IPv4 networking)  
ADD SCTPHOST: SCTPHOSTID=1, IPVERSION=IPv4, SIGIP1V4="x.x.x.x", SIGIP1SECSWITCH=DISABLE,  
SIGIP2SECSWITCH=DISABLE, PN=36422, SCTPTEMPLATEID=0;  
//Adding an X2 control-plane host (in IPv6 networking)  
ADD SCTPHOST: SCTPHOSTID=1, IPVERSION=IPv6, SIGIP1V6="xx:xx:xx:xx", SIGIP1SECSWITCH=DISABLE,  
SIGIP2SECSWITCH=DISABLE, PN=36422, SCTPTEMPLATEID=0;  
//Adding an X2 control-plane peer (in IPv4 networking)  
ADD SCTPPEER: SCTPPEERID=1, IPVERSION=IPv4, SIGIP1V4="x.x.x.x", PN=36422;  
//Adding an X2 control-plane peer (in IPv6 networking)  
ADD SCTPPEER: SCTPPEERID=1, IPVERSION=IPv6, SIGIP1V6="xx:xx:xx:xx", PN=36422;  
//Adding an X2 user-plane host (in IPv4 networking)  
ADD USERPLANEHOST: UPHOSTID=1, IPVERSION=IPv4, LOCIPV4="x.x.x.x", IPSECSWITCH=DISABLE;  
//Adding an X2 user-plane host (in IPv6 networking)
```

```

ADD USERPLANEHOST: UPHOSTID=1, IPVERSION=IPv6, LOCIPV6="xx::xx:xx:xx", IPSECSWITCH=DISABLE;
//Adding an X2 user-plane peer (in IPv4 networking)
ADD USERPLANEPEER: UPPEERID=1, IPVERSION=IPv4, PEERIPV4="x.x.x.x", IPSECSWITCH=DISABLE;
//Adding an X2 user-plane peer (in IPv6 networking)
ADD USERPLANEPEER: UPPEERID=1, IPVERSION=IPv6, PEERIPV6="xx::xx:xx:xx", IPSECSWITCH=DISABLE;
//Adding the X2 user-plane host to the endpoint group
ADD UPHOST2EPGRP: EPGROUPID=1, UPHOSTID=1;
//Adding the X2 user-plane peer to the endpoint group
ADD UPPEER2EPGRP: EPGROUPID=1, UPPEERID=1;
//Adding the X2 control-plane host to the endpoint group
ADD SCTPHOST2EPGRP: EPGROUPID=1, SCTPHOSTID=1;
//Adding the X2 control-plane peer to the endpoint group
ADD SCTPPEER2EPGRP: EPGROUPID=1, SCTPPEERID=1;
//Adding an X2 object on the gNodeB side
ADD GNBCUX2: gNBCuX2Id=1, CpEpGroupId=1, UpEpGroupId=1;
//Besides preceding X2 configurations, the following X2 configurations are required in dual-MPT networking
//Old model
//For each of the two local IP addresses of the X2 interface, configuring an IP route from the eNodeB to the
gNodeB (The two IP addresses of the gNodeB are in the network segment of the route.)
ADD SRCIPRT: SRCRTIDX=0, SN=6, SBT=BASE_BOARD, SRCIP="x.x.x.x", RTTYPE=NEXTHOP,
NEXTHOP="x.x.x.x";
ADD SRCIPRT: SRCRTIDX=1, SN=7, SBT=BASE_BOARD, SRCIP="x.x.x.x", RTTYPE=NEXTHOP,
NEXTHOP="x.x.x.x";
//New model
//For each of the two local IP addresses of the X2 interface, configuring an IP route from the eNodeB to the
gNodeB (The two IP addresses of the gNodeB are in the network segment of the route.)
ADD SRCIPROUTE4: SRCRTIDX=0, SRCIP="x.x.x.x", RTTYPE=NEXTHOP, NEXTHOP="x.x.x.x";
ADD SRCIPROUTE4: SRCRTIDX=1, SRCIP="x.x.x.x", RTTYPE=NEXTHOP, NEXTHOP="x.x.x.x";
//Adding a control-plane host for the X2 interface (Two local IP addresses must be configured for the SCTP
host.)
ADD SCTPHOST: SCTPHOSTID=0, IPVERSION=IPv4, SIGIPV4="x.x.x.x", SIGIP1SECSWITCH=DISABLE,
SIGIP2V4="x.x.x.x", SIGIP2SECSWITCH=DISABLE, PN=36422, SIMPLEMODESWITCH=SIMPLE_MODE_OFF,
SCTPTEMPLATEID=0;
//Adding user-plane hosts for the X2 interface (Two user-plane hosts must be configured.)
ADD USERPLANEHOST: UPHOSTID=1, IPVERSION=IPv4, LOCIPV4="x.x.x.x", IPSECSWITCH=DISABLE,
FLAG=MASTER;
ADD USERPLANEHOST: UPHOSTID=10, IPVERSION=IPv4, LOCIPV4="x.x.x.x", IPSECSWITCH=DISABLE,
FLAG=MASTER;

//Adding the two X2 user-plane hosts to the endpoint group
ADD UPHOST2EPGRP: EPGROUPID=1, UPHOSTID=1;
ADD UPHOST2EPGRP: EPGROUPID=1, UPHOSTID=10;

```

Configuring parameters on the gNodeB (IPv4/IPv6 dual-stack transmission)

```

//Turning on the SA X2 setup switch
MOD GNBINTERFACEPARAM: X2ConfigSw=X2_SETUP_FOR_SA_SW-1;
//Configuring common parameters
//Optional Adding an SCTP template. By default, the gNodeB has an SCTP template with
SCTPTEMPLATEID set to 0.
ADD SCTPTEMPLATE: SCTPTEMPLATEID=0, SWITCHBACKFLAG=ENABLE;
//Adding an endpoint group for the X2 interface
ADD EPGROUP: EPGROUPID=1, IPVERPREFERENCE=IPv6;
//Adding X2 control-plane hosts
ADD SCTPHOST: SCTPHOSTID=1, IPVERSION=IPv4, SIGIPV4="x.x.x.x", SIGIP1SECSWITCH=DISABLE,
SIGIP2SECSWITCH=DISABLE, PN=36422, SCTPTEMPLATEID=0;
ADD SCTPHOST: SCTPHOSTID=2, IPVERSION=IPv6, SIGIPV6="xx::xx:xx:xx", SIGIP1SECSWITCH=DISABLE,
SIGIP2SECSWITCH=DISABLE, PN=36422, SCTPTEMPLATEID=0;
//Required for manual X2 control-plane peer configuration Adding an X2 control-plane peer through
either of the following configurations
ADD SCTPPEER: SCTPPEERID=1, IPVERSION=IPv4, SIGIPV4="x.x.x.x", PN=36422;
ADD SCTPPEER: SCTPPEERID=2, IPVERSION=IPv6, SIGIPV6="xx::xx:xx:xx", PN=36422;
//Adding X2 user-plane hosts
ADD USERPLANEHOST: UPHOSTID=1, IPVERSION=IPv4, LOCIPV4="x.x.x.x", IPSECSWITCH=DISABLE;
ADD USERPLANEHOST: UPHOSTID=2, IPVERSION=IPv6, LOCIPV6="xx::xx:xx:xx", IPSECSWITCH=DISABLE;
//Required for manual X2 user-plane peer configuration Adding an X2 user-plane peer through either of
the following configurations (It is recommended that the IP version on the user plane be the same as that
on the control plane.)
ADD USERPLANEPEER: UPPEERID=1, IPVERSION=IPv4, PEERIPV4="x.x.x.x", IPSECSWITCH=DISABLE;
ADD USERPLANEPEER: UPPEERID=2, IPVERSION=IPv6, PEERIPV6="xx::xx:xx:xx", IPSECSWITCH=DISABLE;

```

```
//Adding the X2 user-plane hosts to the endpoint group
ADD UPHOST2EPGRP: EPGROUPID=1, UPHOSTID=1;
ADD UPHOST2EPGRP: EPGROUPID=1, UPHOSTID=2;
//(Required for manual X2 user-plane peer configuration) Adding the X2 user-plane peer to the endpoint
group through either of the following configurations
ADD UPPEER2EPGRP: EPGROUPID=1, UPPEERID=1;
ADD UPPEER2EPGRP: EPGROUPID=1, UPPEERID=2;
//Adding the X2 control-plane hosts to the endpoint group
ADD SCTPHOST2EPGRP: EPGROUPID=1, SCTPHOSTID=1;
ADD SCTPHOST2EPGRP: EPGROUPID=1, SCTPHOSTID=2;
//(Required for manual X2 control-plane peer configuration) Adding the X2 control-plane peer to the
endpoint group through either of the following configurations
ADD SCTPPEER2EPGRP: EPGROUPID=1, SCTPPEERID=1;
ADD SCTPPEER2EPGRP: EPGROUPID=1, SCTPPEERID=2;
//Adding an X2 object on the gNodeB side
ADD GNBCUX2: gNBCuX2Id=1, CpEpGroupId=1, UpEpGroupId=1;
```

X2 Self-Removal

Configuring parameters on the eNodeB

```
//Activating X2 self-removal triggered by link faults
MOD ENODEBALGOEXTSWITCH: X2SonDelTimerForLNX2Fault=10080;
//Activating X2 self-removal based on the initial setup state
MOD ENODEBALGOEXTSWITCH: GnbX2InitFailDelSwitch=ON;
//Activating X2 self-removal based on X2 usage
MOD ENODEBALGOEXTSWITCH: X2SonDelTimerForLNX2Usage=10080, X2SonDeleteGnbAddCntThld=0;
```

Configuring parameters on the gNodeB

```
//Activating X2 self-removal triggered by link faults
MOD GNBX2SONCONFIG: X2SonConfigSwitch=X2SON_DEL_FOR_X2FAULT_SWITCH-1,
X2SonDeleteTimerForX2Fault=10080;
//Activating X2 self-removal triggered by immediate faults after initial X2 self-setup
MOD GNBX2SONCONFIG: X2SonConfigSwitch=X2INIT_FAIL_DEL_SWITCH-1;
//Activating X2 self-removal based on X2 usage
MOD GNBX2SONCONFIG: X2SonConfigSwitch=X2SON_DEL_FOR_X2USAGE_SWITCH-1,
X2SonDeleteGnbAddCntThld=0, X2SonDeleteTimerForX2Usage=10080;
```

X2 Blacklist and Whitelist

Configuring parameters on the eNodeB

```
//Adding a peer gNodeB to the blacklist
ADD ENDCX2BLACKWHITELIST: Mcc="460", Mnc="01", gNodeBId=1, ListType=BLACKLIST;
//Adding a peer gNodeB to the whitelist
ADD ENDCX2BLACKWHITELIST: Mcc="460", Mnc="01", gNodeBId=2, ListType=WHITELIST;
//Removing peer gNodeBs from the blacklist/whitelist
RMV ENDCX2BLACKWHITELIST: Mcc="460", Mnc="01", gNodeBId=1;
RMV ENDCX2BLACKWHITELIST: Mcc="460", Mnc="01", gNodeBId=2;
```

Other Optimization Functions

Run the following commands to separate X2-U link self-setup/self-removal procedures for different operators in NSA networking.

```
MOD ENODEBALGOEXTSWITCH: NetworkPrfmOptSwitch=SHARE_FREQ_X2_UP_OPT_SW-1;
MOD GNBX2SONCONFIG: X2SonConfigSwitch=X2SON_X2U_AUTO_DEL_OPT_SWITCH-1;
MOD GNBINTERFACEPARAM: InterfaceSelfCfgOptSw=X2_UP_SELF_SETUP_NO_MOCN_SW-1;
```

4.7.1.3 Using the MAE-Deployment

For detailed operations, see Feature Configuration Using the MAE-Deployment.

 NOTE

Due to differences between OSS versions, the GUIs in the interactive video may differ from the actual OSS GUIs and are thus for reference only.

4.7.2 Activation Verification

To check whether an SA X2 link is set up properly, perform the following steps:

- On the LTE side, run the **DSP X2INTERFACE** command. If the command output shows that the value of **X2Interface state** is **Normal** and the value of **Neighboring Base Station Type** is **SA_ONLY_GNODEB**, the SA X2 link has been set up properly.
- On the NR side, run the **LST GNBOPERATOR** and **DSP GNBCUX2INTERFACE** commands. If the command outputs show that the value of **NR Networking Option** is **Standalone** and the value of **gNodeB CU X2 Interface State** is **Normal**, the SA X2 link has been set up properly.

4.7.3 Network Monitoring

Counter ID	Counter Name
1526727202	L.Sig.X2.SendSetup.Att
1526727203	L.Sig.X2.SendSetup.Succ
1526728766	L.X2.Unavail.Dur.Sys
1526729660	L.Signal.Num.X2

5 Parameters

The following hyperlinked EXCEL files of parameter documents match the software version with which this document is released.

- Node Parameter Reference: contains device and transport parameters.
- Node Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- eNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- eNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.
- gNodeBFunction Parameter Reference: contains all parameters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Used Reserved Parameter List: contains the reserved parameters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of parameter reference and used reserved parameter list for the software version used on the live network from the product documentation delivered with that version.

FAQ 1: How do I find the parameters related to a certain feature from parameter reference?

- Step 1** Open the EXCEL file of parameter reference.
- Step 2** On the **Parameter List** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID.
- Step 3** Click **OK**. All parameters related to the feature are displayed.

----End

FAQ 2: How do I find the information about a certain reserved parameter from the used reserved parameter list?

- Step 1** Open the EXCEL file of the used reserved parameter list.

Step 2 On the **Used Reserved Parameter List** sheet, use the **MO**, **Parameter ID**, and **BIT** columns to locate the reserved parameter, which may be only a bit of a parameter. View its information, including the meaning, values, impacts, and product version in which it is activated for use.

----End

6 Counters

The following hyperlinked EXCEL files of performance counter reference match the software version with which this document is released.

- Node Performance Counter Summary: contains device and transport counters.
- eNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- gNodeBFunction Performance Counter Summary: contains all counters related to radio access functions, including air interface management, access control, mobility control, and radio resource management.
- eNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.
- gNodeBFunction Used Reserved Counter List: contains the reserved counters that are in use and those that have been disused.

NOTE

You can find the EXCEL files of performance counter reference for the software version used on the live network from the product documentation delivered with that version.

You can find the EXCEL files of performance counter reference and used reserved counter list for the software version used on the live network from the "Man-Machine Interface Reference" node in the *3900 & 5900 Series Base Station Product Documentation* delivered with that version.

FAQ: How do I find the counters related to a certain feature from performance counter reference?

Step 1 Open the EXCEL file of performance counter reference.

Step 2 On the **Counter Summary(En)** sheet, filter the **Feature ID** column. Click **Text Filters** and choose **Contains**. Enter the feature ID.

Step 3 Click **OK**. All counters related to the feature are displayed.

----End

7 Glossary

For the acronyms, abbreviations, terms, and definitions, see *Glossary*.

8 Reference Documents

- 3GPP TS 37.340 "Evolved Universal Terrestrial Radio Access (E-UTRA) and NR; Multi-connectivity"
- 3GPP TS 36.423 "Evolved Universal Terrestrial Radio Access Network (E-UTRAN); X2 application protocol (X2AP)"
- *S1 and X2 Self-Management*
- *X2 and S1 Self-Management in NSA Networking*
- *Multi-RAT Coordinated Energy Saving*
- *Multi-Operator Sharing*
- *RAN Sharing in eRAN Feature Documentation*